

The Intergenerational Health Effects of Child Marriage Bans ^{*}

Dung D. Le [†] Teresa Molina [‡] Yoko Ibuka [§] Rei Goto [¶]

October 8, 2025

Abstract

Using data from 17 countries, we investigate the effects of child marriage bans on child mortality in the next generation. We use within-country variation in mothers' exposure to the ban across cohorts and in "treatment intensity," calculated based on region-level child marriage prevalence and marriage age prior to the ban. We find that child marriage bans reduced under-5 mortality, with an effect of 19.7 percent corresponding to a one standard deviation change in treatment intensity. Increases in age at first marriage and first birth, which may have led to improved health-related decisions around the time of birth, appear to be the main drivers of the mortality reductions documented.

Keywords: child marriage bans, early marriage, child mortality, Demographic and Health Surveys
JEL classification: I10, I15, J12, J18

^{*}This work has greatly benefited from the feedback of seminar/meeting participants at Japanese Economic Association, the Japan-Taiwan-Korea Health Economics Association Joint Conference, Hitotsubashi University, Sophia University, University of Connecticut, University of Hawaii at Manoa, NBER, Essen Health Conference, and Leibniz University Hannover. Financial support from JSPS Kakenhi (23K20578) is greatly appreciated.

[†]Corresponding author. Faculty of Economics, National Economics University (dungld@neu.edu.vn) and Waseda Institute for Advanced Study (WIAS), Waseda University

[‡]Department of Economics, University of Hawaii at Manoa

[§]Faculty of Economics, Keio University

[¶]Graduate School of Business Administration, Keio University

1 Introduction

There is substantial causal evidence documenting that later marriage yields better education and labor market outcomes for women (Field and Ambrus, 2008; Asadullah and Wahhaj, 2019; Sunder, 2019), as well as better health and education outcomes for their children (Sekhri and Debnath, 2014; Chari et al., 2017; Asadullah and Wahhaj, 2019; Garcia-Hombrados, 2022). Yet, many countries continue to document high rates of child marriage (UNICEF, 2017), for which the existing literature has explored a variety of economic and sociocultural explanations (Corno et al., 2020; Psaki et al., 2021; Buchmann et al., 2023; Corno and Voena, 2023). For governments interested in eliminating the practice of child marriage, age-of-marriage laws are, perhaps, the most direct policy lever available. While many studies have examined the effects of these laws on the outcomes of women (Bergstrom and Özler, 2023), there has been much less research on the effects of these laws on the health of the next generation.

In this paper, we estimate the intergenerational effect of child marriage bans, which set a legal minimum age of marriage to 18, on mortality rates among children of the affected women. Existing work shows that child marriage bans improve women’s socioeconomic outcomes and delay age of marriage in some contexts (McGavock, 2021; Rokicki, 2021; Wilson, 2022). We also know, primarily from studies that use age of menarche as an instrumental variable, that delaying age of marriage leads to better outcomes for children (Sekhri and Debnath, 2014; Chari et al., 2017; Asadullah and Wahhaj, 2019; Sunder, 2019; Garcia-Hombrados, 2022). However, it is not clear whether and to what extent child marriage bans will improve child health, especially given that age of marriage laws are often not properly enforced (Batyra and Pesando, 2021; Collin and Talbot, 2023) and can sometimes lead to substitution away from marriage to informal unions (Bellés-Obrero and Lombardi, 2023). In addition, child marriage bans may have the unintended effect of making child brides more hesitant to seek prenatal or postnatal care (for fear of legal punishment), which could lead to worse health outcomes for their children. In short, it is not clear whether child marriage bans will generate health benefits for the next generation.¹

For this analysis, we use the MACHEquity Child Marriage Policy Database, which contains information on child marriage bans over the period 1995 to 2012, to identify the timing of the bans in each country (World Policy Center, 2022). This country-level dataset is then linked to the

¹Three papers investigate an Ethiopian reform that raised the legal minimum age of marriage and reach different conclusions about the effects of this policy on infant mortality (McGavock, 2021; Rokicki, 2021; Garcia-Hombrados, 2022).

Demographic and Health Surveys (DHS), representative surveys of women aged 15-49 in low- and middle-income countries (LMICs), from which the main outcomes are extracted. Our final dataset includes 17 countries that legally banned child marriage under the age of 18 during the period from 1995 to 2012.

To isolate the causal effects of the bans, we rely on a similar strategy as [Wilson \(2022\)](#), which uses the same data to analyze the effect of child marriage bans on women’s socioeconomic outcomes. This strategy exploits two sources of variation: subnational regional differences in the pre-ban age at marriage and variation across cohorts within countries in exposure to the bans. We first calculate a region-specific measure of treatment intensity, defined such that locations where, in the pre-ban period, the occurrence of child marriage was common and child brides married particularly young are considered to have high treatment intensity, which means that individuals in these locations should be more affected by a child marriage ban.² The treatment intensity variable is then interacted with an indicator for cohorts exposed to the bans (i.e., individuals under the age of 18 at the time a ban was implemented in their country). If the bans reduce child mortality, then there should be a larger decrease from women who turned 18 before to those who turned 18 after a ban in areas with high treatment intensity compared to areas of low treatment intensity.

Our results show that the bans have statistically significant and sizable effects on child mortality. Specifically, our main specification reveals a one standard deviation (SD) increase in treatment intensity reduced infant and under-5 mortality by about 0.95 and 1.97 percentage points, respectively (approximately 14.2 and 19.7 percent relative to the mean of pre-ban cohorts). Effects were driven by low-income countries. For under-5 mortality, the pattern of our coefficients from a standard event study analysis, as well as the approach proposed by [Callaway et al. \(2024\)](#), confirms our findings and lends support to the validity of our empirical strategy. For infant mortality, evidence from these alternative specifications is much weaker. Our main regression estimates are robust to various specification tests, including those that account for contemporaneous policies, differential trends due to regional characteristics, alternative samples, and migration. We thoroughly explore issues relating to selection into motherhood.

The bans appear to be reducing child mortality primarily by delaying age at first marriage and first birth. Women going into marriage or pregnancy with more maturity should have more agency and bargaining power, which are likely to be important for prenatal and postnatal health

²This approach is similar to many other generalized difference-in-differences papers ([Bleakley, 2007](#); [Cutler et al., 2010](#); [Lucas, 2010](#); [Barofsky et al., 2015](#); [Wilson, 2022](#)).

investments. Although this information is only available for a subset of our original sample, we document suggestive evidence that child marriage bans affect mothers’ health-seeking decisions around the time of birth (specifically, place of delivery and child vaccinations).

Our paper contributes to a large body of evidence on the impacts of child marriage laws on the outcomes of women and their children. While most work has focused on the impacts on women – age at marriage, education, and economic outcomes (Dahl, 2010; Bharadwaj, 2015; Wang and Wang, 2017; McGavock, 2021; Wilson, 2022; Bellés-Obrero and Lombardi, 2023; Collin and Talbot, 2023) – our study aims to shed light on the impacts on child mortality, for which existing evidence is scant, mixed, and currently only focused on Ethiopia (McGavock, 2021; Rokicki, 2021; Garcia-Hombrados, 2022). A key distinguishing feature of our study, compared to these three, is its scope: our sample includes 17 developing countries across four continents. In addition to providing more external validity, the use of pooled data from multiple countries helps reduce the possibility of biased estimates due to the low rates of infant and under-5 mortality, as discussed by Dursun et al. (2017).³ This also allows us to investigate sources of heterogeneity and potential underlying mechanisms.

This study is also closely related to a well-established literature on the effects of delaying marriage (outside of the context of child marriage laws). Studies in this literature typically use age at menarche as an instrumental variable for age at marriage (Field and Ambrus, 2008; Sekhri and Debnath, 2014; Chari et al., 2017; Asadullah and Wahhaj, 2019; Sunder, 2019; Garcia-Hombrados, 2022; Tauseef and Sufian, 2024). Because age at menarche may not satisfy the exclusion restriction for a number of reasons, our paper offers valuable evidence on this topic by using a different source of variation.⁴

Child marriage bans are just one example of an intervention aimed at reducing child marriage, and our study therefore speaks to a broader literature on this more general class of interventions. Bergstrom and Özler (2023) review studies on 15 categories of interventions (including age of marriage laws, job opportunities, and cash transfers) and conclude that none have been consis-

³It has been suggested that in the case of rare events (e.g., infant and child mortality) where the mean value is small, estimates from a small sample size could result in bias. Specifically, Dursun et al. (2017), using three waves of the Turkish DHS, show that small sample sizes with low average values of child mortality and low birth weight lead to biased estimates in their study. In addition, Chari et al. (2017) and McGavock (2021) argue that low child mortality rates tend to produce noisy estimates due to low statistical power.

⁴Huang et al. (2019) describe how menarcheal age may be correlated with cognitive development trajectories, early exposure to family-related disruptions, stress, and malnutrition, which can generate violations of the exclusion restriction if appropriate controls are not available. Wilson (2022) suggests that selection in the marriage market may also lead to violations.

tently found to be effective at reducing child marriage. Nevertheless, several recent papers have documented that community-based education and economic incentives can reduce child marriage (Chow and Vivalt, 2022; Buchmann et al., 2023; Cohen et al., 2023; Giacobino et al., 2024). The results of our study suggest that the benefits of these interventions could extend to the children of the affected women as well.

2 Background and Data

In this section, we provide some background on child marriage across the globe and describe the two sources of data used in our analysis: country-level data on the timing of the bans in each country (MACHEquity Child Marriage Policy Database) and individual-level data on child mortality and related outcomes (the DHS).

2.1 Child Marriage

Since the 1940’s, several international agreements have contributed to forming a general global recognition of the need for action against child marriage.⁵ For example, the 1995 Beijing Declaration and Platform for Action urged all countries to “enact and strictly enforce” a legal minimum age at marriage and reaffirm age 18 as the accepted legal minimum age at marriage. The declaration also called for banning child marriage worldwide (United Nations, 1995). After the Beijing Declaration, the proportion of countries allowing child marriage fell significantly – from 80% in 1995 to 56% in 2013 (Arthur et al., 2018). Eradicating child marriage is now one of the main targets of the fifth Sustainable Development Goal (Wodon et al., 2017).

Despite international efforts, child marriage still takes place across many countries, cultures, religions, and ethnicities. Child marriage is much more prominent among girls than boys, and it is most common in South Asia and sub-Saharan Africa. Globally, it is estimated that 29% of women aged 20 to 49 were married before the age of 18. In South Asia, this figure is 56% (UNICEF, 2017).

⁵See Wilson (2022) for a more detailed description of these agreements.

2.2 MACHEquity Child Marriage Policy Database

We use the MACHEquity child marriage policy database to identify the timing of child marriage bans in each country. This database was constructed using various reliable legal sources including country government websites, the Foreign Law Guide, the Lexadin World Law Guide, and the NATLEX database, among others (World Policy Center, 2022). It contains detailed information on the minimum legal age of marriage with parental consent and minimum legal age without parental consent from 1995 to 2012 for both males and females. The minimum age at marriage in each country is categorized into five groups: “18 years or older”; “16 or 17 years old”; “14 or 15 years old”; “9 to 13 years old”; and “No minimum age.” Because allowing child marriage without parental consent is not common and because child marriage is more prevalent among females, we follow Wilson (2022) and focus on the legal minimum age at marriage with parental consent for females. We further restrict the database to countries that have information on the legal minimum age at marriage with parental consent in 1995, resulting in a balanced panel of 105 countries. Of these countries, there are 18 countries which changed their legal minimum age at marriage with parental consent to 18 during the 1995-2012 period and also have a DHS survey following this law change. Appendix Table A1 lists all of these countries, along with the year of the child marriage ban and the most recent survey year of the DHS. The earliest countries to ban child marriage were Kazakhstan and Peru in 1998 and 1999, respectively, while Liberia banned child marriage most recently in 2012. There is substantial variation in the time between ban year and survey year, ranging from 1 to 17 years. We exclude Kazakhstan from our analysis because only ten mothers surveyed in the DHS were born recently enough to have been exposed to the ban.

2.3 Demographic and Health Survey (DHS) data

We extract child mortality and other outcomes from the DHS, a series of health-focused household surveys administered to women of childbearing age across a wide range of lower and middle income countries (ICF, 2018). For countries with multiple DHS surveys fielded after the implementation of a child marriage ban, we use the most recent survey available.

The DHS uses a stratified two-stage cluster design. Specifically, it first draws enumeration areas from national census files and then draws a sample of households from each enumeration area. Over the past three decades, the DHS has asked a comparable set of questions to representative samples of women aged 15-49. Because we are interested in child health, we restrict our attention to

women who report having ever given birth. We also drop any women who themselves were born after a child marriage ban was implemented in their country to reduce the likelihood of picking up indirect effects operating through their mother’s exposure to the ban. This restriction only drops 25 women, leaving us with a sample size of 200,581 women from 17 countries, with an average age of 24.2 at the time of the ban and 32.8 at the time of survey, as reported in column 1 of Table 1. 42% of the sample lives in an urban area.

Although each survey round is a cross-section, the survey collects a retrospective birth history that documents information on each child ever born alive to the respondent. The birth history module records detailed information on the month of birth of the child, whether he or she died and the age in months at death. These questions are the same across survey rounds and countries, making the child death measures comparable across countries.

Using the birth history and reported age at death, we construct indicators of infant mortality (death between birth and the first birthday) and under-5 mortality (death between birth and the fifth birthday), two outcomes that are widely used in economic studies of child death (Kuecken et al., 2021; Chou et al., 2010; Lu and Vogl, 2023; Kammerlander and Schulze, 2023; Baird et al., 2011; Le et al., 2025) and documented in international reports (United Nations Inter-agency Group for Child Mortality Estimation, 2018).

We focus on mortality of the first-born child. If child marriage bans affect the number of children primarily on the intensive margin but not the extensive margin (which is consistent with what we find), compositional changes in the higher-parity births that we observe would make it difficult to interpret estimates for higher-parity births. Approximately 6.3% and 9.4% of women have experienced the death of their first-born child under age 1 and age 5, respectively.

To explore potential mechanisms, we also utilize information on our sample mothers’ educational attainment, employment, age at first marriage, age at first birth, and total fertility. As we report in column 1 of Appendix Table A2, the mothers in our sample married and gave birth young (on average at ages 18.9 and 20.1, respectively), with 29% of them having given birth before turning 18. Educational attainment (5.7 years) and employment rates (0.58) are low. At the time of survey, mothers in the sample have an average of 3.6 children – a high number given that the respondents themselves are 32.8 years old on average. We also make use of information relating to prenatal, at-birth, and postnatal healthcare, along with measures of child health (height-for-age, weight-for-age, and weight-for-height). Unfortunately, these variables are only available for a

subset of our original sample, as we detail in Appendix Table A3.

Table 1: Summary Statistics

	(1) Unweighted	(2) Weighted
Age at survey	32.8 (8.52)	32.9 (8.48)
Age at ban	24.2 (9.89)	24.3 (10.1)
Urban	0.42 (0.49)	0.42 (0.49)
Female child	0.48 (0.50)	0.48 (0.50)
Infant Mortality	0.063 (0.24)	0.064 (0.25)
Under-5 Mortality	0.094 (0.29)	0.096 (0.29)
Observations	200581	200581

Notes: Standard deviations in parentheses. Sample restricts to women who have ever given birth and who were born before the child marriage ban was implemented in their country. Column 2 weights the sample using the methods described in section 3.3.

3 Empirical Strategy

3.1 Generalized Difference-in-Differences

Countries that implement child marriage bans are systematically different in various ways from countries that do not; often, a ban is implemented as a response to very high rates of child marriage. Cross-country comparisons would therefore fail to provide a causal estimate of the effect of child marriage bans on child health. For this reason, our entire analysis restricts to countries that did implement a child marriage ban (sometime between 1995 and 2012) and compares the child mortality rates of affected women with those of unaffected women. Women under 18 at the time of a ban’s implementation would have been affected by the ban, whereas those 18 and older would have not. Simply comparing the children of affected and unaffected cohorts, however, does not allow us to isolate the effect of the ban from other trends in child health across maternal birth cohorts. We therefore also take advantage of within-country variation in how consequential a ban would actually be, given existing marriage practices. In sum, our empirical strategy relies on two

sources of variation: (1) within-country variation in exposure to the bans across cohorts, and (2) within-country variation in “treatment” intensity across regions, where our treatment of interest is a child marriage ban. A region is defined by the largest geographic administrative region in the country, along with the urban-rural status of the exact location. For example, urban areas of one state will form one region, while rural areas of that same state will form a separate region.

Following [Wilson \(2022\)](#), we define treatment intensity in subnational region r , focusing on “pre-ban” cohorts aged 18 to 30 at the time of the ban, as follows:⁶

$$\text{intensity}_r = \frac{\sum_{i=1}^{N_r} \mathbb{1}(\text{married before 18})_i \times (18 - \text{age first married}_i)}{N_r} \quad (1)$$

The denominator (N_r) is the number of women in region r who belong to these pre-ban cohorts, unaffected by the ban because they were at least 18 years old when the ban was implemented in their country. The numerator is the summation (across all N_r women in pre-ban cohorts in region r) of the product of two terms: an indicator for getting married before the age of 18 (extensive margin) and the difference between 18 and the woman’s age at first marriage (intensive margin). This definition treats a ban as having greater intensity in subnational regions where the pre-ban average age at marriage is lower. The idea is that a ban should have stronger effects where, in the absence of the ban, women were marrying at younger ages. Defining treatment intensity in this way takes advantage of considerable regional variation in the prevalence of child marriage pre-ban. Figure 1 exhibits the distribution of our treatment intensity variable, which has a mean of 1.27 and a standard deviation of 0.72. Appendix Figure A1 displays a separate histogram for each country.

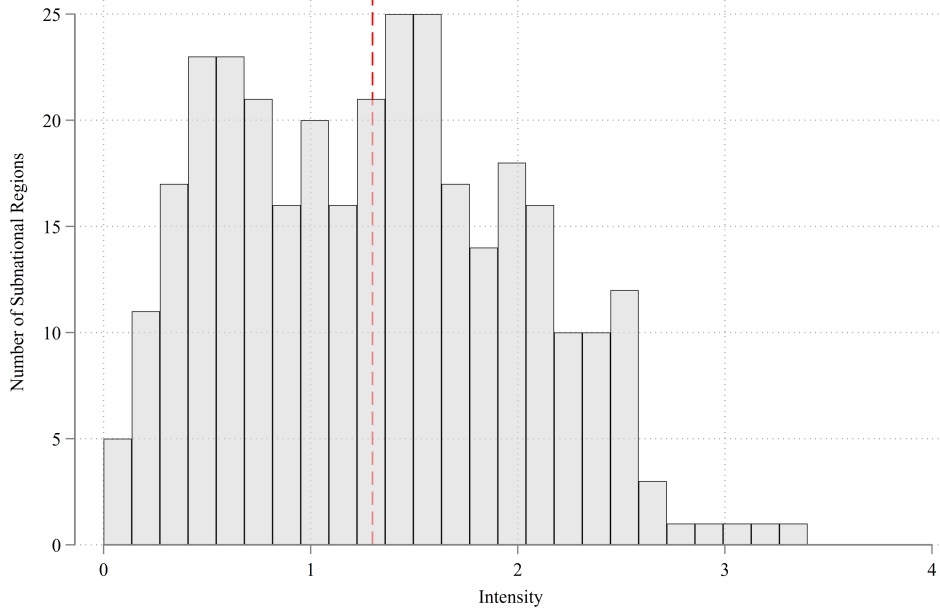
To estimate the effect of child marriage bans on child mortality of the next generation, we restrict to the first-born child of all mothers in the DHS who were born before the child marriage ban was implemented in their country. We estimate the following specification:

$$y_{ikr} = \alpha(\text{postban}_{kc(r)} \times \text{intensity}_r) + \gamma_{kc(r)} + \theta_r + \epsilon_{ikr}, \quad (2)$$

where y_{ikr} represents the outcome of interest (firstborn mortality indicators or a related potential mechanism) for mother i in cohort k in subnational region r , which is part of a country that we

⁶While the upper age limit of 30 ensures that we are calculating intensity using a similar age-at-ban distribution across countries, dropping this restriction generates a variable that is highly correlated ($r=0.96$) with our current intensity variable and results that are very similar to our current set of results.

Figure 1: Distribution of Treatment Intensity Variable



Notes: Treatment intensity is calculated as described in Equation (1). The vertical dashed line depicts the mean (1.27). The standard deviation is 0.72

denote using subscript c . $postban_{kc(r)}$ is an indicator variable coded as one if cohort k in country c is under the age of 18 at the time of the of the ban (the affected post-ban cohorts) and zero otherwise (unaffected pre-ban cohorts). We are interested in α , the coefficient on the interaction between $intensity_r$ and $postban_{kc(r)}$, which captures how the difference in child mortality between affected and unaffected cohorts varies across areas with different treatment intensity. Like Wilson (2022), we control for both $\gamma_{kc(r)}$ and θ_r . Country-specific cohort effects ($\gamma_{kc(r)}$) account for factors like age at the time of survey and country-specific differences in marriage or fertility patterns across cohorts. Subnational region fixed effects (θ_r) control for time-invariant unobserved characteristics specific to each region, including geographic and cultural factors. In all estimations, we cluster standard errors at the subnational region level to account for potential correlations of error terms within regions.⁷

The validity of this empirical strategy rests on the assumption that, conditional on time-invariant subnational region fixed effects and country-specific cohort fixed effects, the relationship between treatment intensity and outcomes would have remained the same for cohorts who were 18 and older and cohorts under 18 at the time of ban, if the ban had not been implemented.

⁷In total, our study sample consists of 328 subnational regions.

3.2 Event Study

One way to shed light on this assumption is to estimate event study regressions that will reveal whether the relationship between treatment intensity and child mortality was changing across cohorts among the set of pre-ban cohorts (none of whom were exposed to the ban). If this is the case, it would suggest the relationship between treatment intensity and mortality was being affected by something other than the implementation of a child marriage ban in a country, and therefore signal a potential violation of the identifying assumption.

In our event study regressions, we replace the post-ban indicator with a set of dummies representing the respondent’s age at the time of the ban, as described in the following specification:

$$y_{ikr} = \sum_{t=-15, t \neq 0}^{14} \alpha_t (\mathbb{1}(\text{age at ban}_{kc(r)} = 18 + t) \times \text{intensity}_r) + \gamma_{kc(r)} + \theta_r + \epsilon_{ikr}. \quad (3)$$

Here, $\mathbb{1}(\text{age at ban}_{kc(r)} = 18 + t)$ refers to a series of dummies indicating the respondent’s age at the time of the ban, ranging from 3 through 32, which creates 15 pre-ban and 15 post-ban cohorts.⁸ $\mathbb{1}(\text{age at ban}_{kc(r)} = 18)$ is omitted as the reference cohort. The set of α_t coefficients are the main coefficients of interest. Since women aged 18 and older at the time of a ban should not be affected by the ban, we expect the coefficients for these cohorts to be statistically insignificant, indicating no differential pre-trends across the treatment intensity distribution for the unaffected cohorts who turned 18 prior to the bans.

A growing literature highlights that the coefficients from Equation (3), essentially a two-way fixed effects regression with staggered treatment adoption, might be difficult to interpret if treatment effects are heterogeneous across regions (Roth et al., 2023). Staggered adoption here refers to the fact that the intensity variable only becomes relevant for cohorts born after a certain year, which depends on when a child marriage ban was implemented in a specific country. Callaway and Sant’Anna (2021) propose methods for estimating the dynamic causal effects of a treatment with staggered adoption across multiple periods. Because these methods were designed for a binary treatment, we use extensions proposed by Callaway et al. (2024) to apply their methods to our continuous treatment setting.

This approach essentially boils down to the use of Callaway and Sant’Anna (2021) methods

⁸We combine those aged 3 and younger at the time of the ban and combine all cohorts aged 32 and older at the time of the ban.

for multiple different binary treatment variables. We estimate two sets of event study treatment effects – one for a “low intensity” treated group and one for a “high intensity” treated group, both of which involve using the same “never-treated” group as a control. The estimation of the low-intensity treatment effects do not use any high intensity observations, and the estimation of the high-intensity treatment effects ignores all low intensity observations.

Strictly speaking, we do not have any never-treated regions in our sample because we restrict to countries that implemented a child marriage ban. However, regions where child marriage was extremely uncommon before the ban can be thought of as essentially untreated. We therefore define those with treatment intensity below 0.23 – the 5th percentile of the region-level treatment intensity distribution – as never-treated.⁹ To define low and high intensity, we split the remaining regions at the 75th percentile (1.8). For computational efficiency, we collapse our data to the region-cohort level, averaging the outcome variables and summing the weights we use in all analyses, described in the next sub-section.

3.3 Weighting Procedure

Because we are interested in child health, our analysis necessarily restricts to women who have ever given birth. It is therefore important to determine whether our variable of interest ($postban_{kc(r)} \times intensity_r$) predicts selection into the sample. If it does, this would imply that a significant α in Equation (2) could be due to changes in the composition of our mother sample – specifically, differences between mothers in the affected versus unaffected cohorts varying across the treatment intensity distribution. This could happen if child marriage bans affected the extensive margin of the fertility decision (most plausibly by reducing the likelihood of a woman ever giving birth). Alternatively, because the DHS surveys women of different ages, differences in the timing of first birth across the treatment intensity distribution could also result in sample composition issues.

To investigate further, we examine all women in the DHS, including those who have never given birth. Our outcome of interest is an indicator variable for having ever given birth (which is equal to 1 for all mothers in our main analysis sample). We then estimate Equation (2) using this indicator as our outcome variable and report our results in column 1 of Table 2. The positive and statistically

⁹A region where 23% of women get married at age 17 would generate an intensity variable of 0.23 (though there are of course many other scenarios that could result in this intensity level). A child marriage ban is unlikely to have much of an effect in a region like this, where the vast majority of women are marrying as adults and the women who do marry as children are close to 18.

significant interaction coefficient reveals that our variable of interest does indeed predict selection into the sample. The sign of the coefficient helps shed light on why. It is unlikely that the bans are actually inducing women (who otherwise would not have had any children) to have children, which is what would be implied by a positive coefficient. Rather, we argue the positive coefficient stems from the fact that the DHS captures women at different ages.¹⁰ The likelihood of being a mother (and therefore being in our sample) is lower for women who are younger at the time of survey, and post-ban women (in later birth cohorts) are younger than pre-ban women at the time of survey. Because areas with high treatment intensity are places where women traditionally married early relative to low intensity areas, post-ban women in high intensity areas are more likely to have already married and given birth by the time they are interviewed and are therefore more likely to be included in our mother sample. In pre-ban cohorts, by contrast, where average age at survey is much higher, the vast majority of women across the treatment intensity distribution have already had a child.

Table 2: Selection

	(1)	(2)	(3)
	In Mother Sample	In Mother Sample	In Mother Sample
post-ban $\times$ intensity	0.0830*** (0.0100)	0.0154 (0.0145)	-0.0008 (0.0111)
Added Survey Age Controls	None	Linear Age $\times$ Intensity	Age Dummies $\times$ Intensity
Mean outcome	0.74	0.74	0.74
Observations	270,315	270,315	270,315

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Sample includes all women who were born before the child marriage ban was implemented in their country. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications control for country-cohort fixed effects and subnational region fixed effects.

To test whether this does indeed explain the positive selection coefficient, we repeat the selection regression described above and allow for trends in motherhood by age at survey to vary across the treatment intensity distribution. In column 2, we do this by controlling for an age-by-intensity interaction, and in column 3, we control for age fixed effects interacted with the intensity variable (both of which we are only able to do because we have data from multiple countries surveyed in different years). As shown in columns 2 and 3 of Table 2, the main coefficient of interest ($postban_{kc(r)} \times intensity_r$) is no longer statistically significant and is much smaller in magnitude.

¹⁰If child marriage bans reduced maternal mortality, this could also generate a positive selection coefficient, but as we argue in section 4.5, maternal mortality rates are likely too low for this to be the case.

This supports our explanation for the positive coefficient in column 1: age at survey is an important predictor of selection into the mother sample and trends in motherhood by age at survey vary across regions of different treatment intensity levels.¹¹ We also note that – even after properly accounting for age at survey – there is no evidence that child marriage bans reduce the likelihood of women becoming mothers (the coefficient in column 2 is positive, and the coefficient in column 3 is close to zero).

The fact that there are more post-ban mothers in high intensity regions than in low intensity regions complicates our interpretation of α in Equation (2), which relies essentially on a difference in differences: between children in high intensity and low intensity regions, born to mothers in post-ban vs. pre-ban cohorts. These “additional” mothers gave birth at an earlier age, and this could affect the child mortality difference-in-differences estimates if childbearing age affects child mortality. To deal with this issue, when we estimate Equations (2) and (3), we re-weight our mother sample to equalize the age (at survey) distribution across subnational regions to match the overall age distribution of the mother sample.¹² This is similar in spirit to adding the age-by-intensity controls used in Table 2 without imposing a specific functional form.

Column 2 of Tables 1 and A2 report summary statistics for our mother sample calculated using these weights. By construction, average age at survey is almost identical across the two columns. In addition, the weighted and unweighted samples look very similar in terms of all other variables.

4 Results

4.1 Main Results

Table 3 presents the main results estimated from Equation (2) using the weighted sample. The results show that the bans had statistically significant and sizeable effects on infant and under-5 mortality. Specifically, a one-unit increase in the intensity of the bans significantly reduced infant and under-5 mortality by approximately 1.3 and 2.7 percentage points, respectively. In terms of SD, a one SD (i.e., 0.728 unit) increase in the intensity of the bans corresponds to reductions in

¹¹While our main specification controls for age at survey through the country-cohort fixed effects ($\gamma_{kc(r)}$), these do not allow for survey age trends to vary across subnational regions.

¹²Let N denote the total number of women in the sample, N_a denote the total number of women of age a , and N_r denote the total number of women in region r . For women in region r aged a at the time of survey, a weight equal to $\frac{\frac{1}{N} \sum_N \mathbb{1}(\text{region}=r)}{\frac{1}{N_a} \sum_{N_a} \mathbb{1}(\text{region}=r)}$, which is equivalent to $\frac{\frac{1}{N} \sum_N \mathbb{1}(\text{age}=a)}{\frac{1}{N_r} \sum_{N_r} \mathbb{1}(\text{age}=a)}$, equalizes the age distribution across regions.

infant and under-5 mortality of approximately 0.95 and 1.97 percentage points. These effect sizes are equivalent to about 14.2% and 19.7% reductions in infant and under-5 mortality relative to the mean of the unaffected pre-ban group.

Table 3: Effects of Child Marriage Bans on First-born Child Mortality

	(1)	(2)
	Infant Mortality	Under-5 Mortality
post-ban $\times$ intensity	-0.013*** (0.003)	-0.027*** (0.005)
Mean outcome of pre-ban cohorts	0.067	0.100
Observations	200,581	200,581

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the ban. All specifications use the weights described in section 3.3 and control for country-cohort fixed effects and subnational region fixed effects.

The coefficients in Table 3 show that treatment intensity, which was positively correlated with infant and under-5 mortality for pre-ban cohorts, became less positively correlated for cohorts young enough to be affected by a child marriage ban, which we attribute to high treatment intensity regions seeing larger reductions in child mortality due to greater “exposure” to the bans. While we have interpreted this as evidence that child marriage bans decreased child mortality, an event study regression can add further support by isolating exactly when the intensity-mortality correlation began to change.

Figure 2 provides a graphical illustration of our event study results. The x-axis represents the respondent’s age at the time of the ban. For each age, we report the regression coefficients along with 95% confidence intervals estimated from Equation (3), which represent the change in the relationship between treatment intensity and child mortality for a particular cohort, relative to the reference cohort group. The dashed vertical line marks this reference group – those aged 18 at the time of the ban. Since the bans only impose restrictions on individuals under the age of 18, the relationship between treatment intensity and child mortality should not vary across cohorts for those aged 18 and older at the time of the bans. Consistent with this, the infant mortality figure reveals a flat pattern of (mostly) statistically insignificant estimates for cohorts aged over 18 who should not have been affected by the bans. The pre-trend is relatively flat for under-5 mortality as well, though it does exhibit a slight downward trend. When we use the methods of Callaway et al.

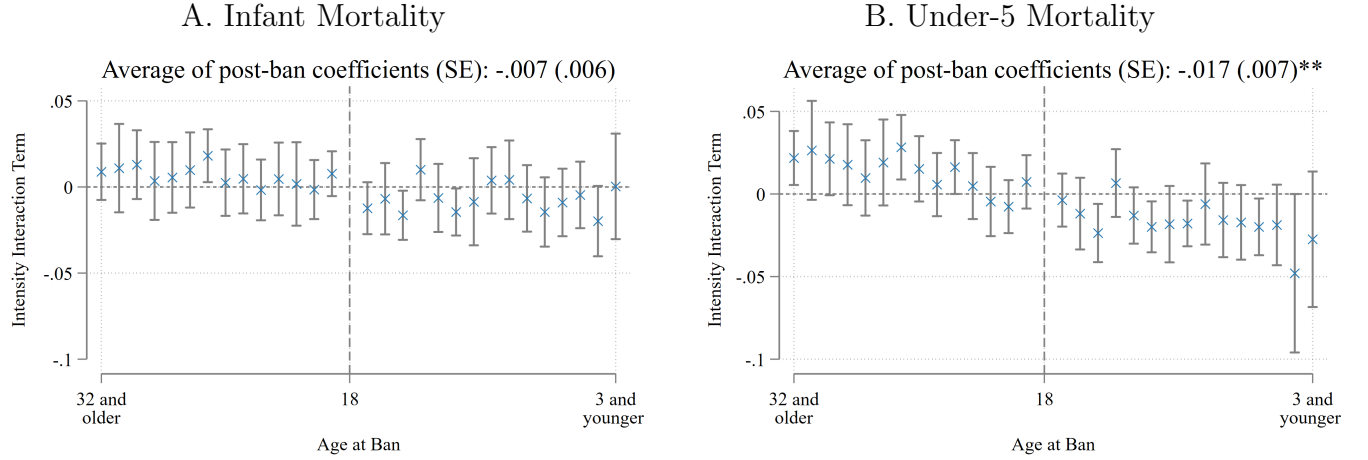
(2024), pre-trends are flat for both outcome variables (Figure 3). These results lend support to the assumption that the relationship between treatment intensity and child mortality would have remained the same if the bans had not been implemented, and that the change in this relationship (summarized by the coefficient estimates in Table 3) is indeed due to the child marriage bans.

As for the ban-affected cohorts, the under-5 mortality results in Figure 2B show a clear shift downwards. With only one exception, the coefficients for post-ban cohorts are negative and several are statistically significant. Although the coefficients for cohorts aged 16 and 17 at the time of the ban are small in magnitude, the treatment intensity interactions become more negative and statistically significant for younger cohorts, and the average of all post-ban coefficients is negative and statistically significant. The larger effect sizes for younger cohorts likely reflect the fact that women who are older but still under the age of 18 at the time of the bans are less exposed to the ban than younger cohorts – either because they are more likely to already be married by the time the ban is implemented or because there is a lag in both public knowledge of the bans and their enforcement.¹³ For infant mortality, the pattern is similar but weaker. Most (but not all) post-ban coefficients are negative, though the coefficients are smaller in magnitude and the coefficients are less precisely estimated than those in the under-5 mortality figure. Unlike in the under-5 figure, the average of the post-ban coefficients for infant mortality is not statistically significant.

In Figure 3, which uses the Callaway et al. (2024) approach, the results for under-5 mortality are consistent with our regression and event study results. As mentioned above, they in fact provide even stronger support for the absence of significant pre-trends. As for the post-ban treatment effects, the comparison between the coefficients for high intensity and low intensity regions is informative. The post-ban treatment effects for low intensity regions are small in magnitude and not significantly different from zero, but the high intensity regions demonstrate a pattern of negative coefficients that increase in magnitude with age at ban and which, on average, are statistically significant. Though the results for infant mortality also reveal no evidence of pre-trends, the pattern of the post-ban coefficients is much weaker and on average not statistically significant for either group. Taken together, our results provide strong evidence that child marriage bans contributed to a statistically significant reduction under-5 mortality (though the evidence is less clear for infant mortality, a rarer event).

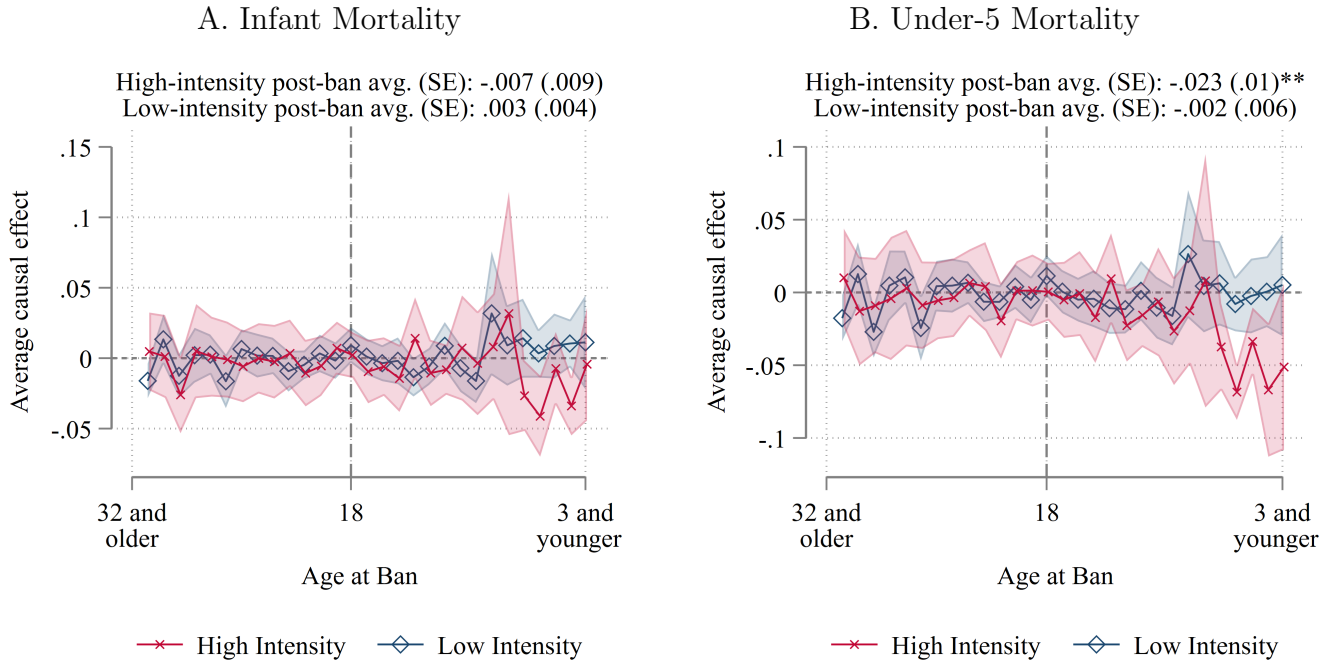
¹³In Ethiopia, for example, there was a staggered rollout of the ban across regions over a ten-year period (McGavock, 2021).

Figure 2: Event Study Results



Notes: Standard errors clustered at region level. Figure reports coefficients (and 95% confidence intervals) on the interactions between treatment intensity and age at ban from Equation (3). The dashed vertical line denotes the omitted reference cohort, those aged 18 at the time of the ban.

Figure 3: Callaway et al. (2024) Results



Notes: Results generated from region-cohort-level data. Figure reports estimates (and 95% confidence intervals) of the average treatment effect for each age at ban, using Callaway and Sant'Anna (2021) methods. We treat regions under the 5th percentile of the treatment distribution as never-treated. Among the remaining treated regions, low intensity regions are those with treatment under the 75th percentile, and high intensity regions are those with treatment greater than or equal to the 75th percentile.

4.2 Heterogeneity Analysis

For which countries and individuals were the effects of these bans the largest? We first estimate heterogeneous effects by country income levels based on World Bank income classifications. Our sample consists of 8 low-income, 4 lower-middle income, and 5 middle-income countries. We find that the negative effect of a child marriage ban diminishes as country income increases: child marriage bans reduced infant and under-5 mortality primarily in low-income and lower-middle countries. As reported in Table 4, coefficient estimates for middle-income countries are statistically insignificant and significantly different from estimates in low-income countries. Estimates for low-income countries are significantly larger than those of lower-middle-income countries. Children in low-income countries, where child marriage, child mortality, and poverty rates are higher (UNFPA-UNICEF, 2019; Wodon et al., 2017), benefit the most from the bans.

Table 4: Heterogeneous effects by country income classification

	Infant Mortality				Under-5 Mortality			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low	Lower-	Middle	Difference	Low	Lower-	Middle	Difference
	Income	Middle	Income		Income	Middle	Income	
		Income				Income		
post-ban $\times$ intensity	-0.028*** (0.007)	-0.010*** (0.003)	-0.004 (0.011)		-0.047*** (0.009)	-0.023*** (0.006)	-0.009 (0.012)	
Low-Mid vs. Low				0.018** (0.007)				0.024** (0.011)
Middle vs. Low				0.023* (0.012)				0.038** (0.015)
Mean outcome of pre-ban cohorts	0.087	0.059	0.040		0.13	0.088	0.054	
Observations	78,927	70,744	50,910	200,581	78,927	70,744	50,910	200,581

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications use the weights described in section 3.3 and control for country-cohort fixed effects and subnational region fixed effects. Country income levels are taken from World Bank income classifications.

Table 5 presents heterogeneity by gender and rural-urban status. In panel A, we show that results are similar for boys and girls. Though coefficient estimates are larger for boys, they are not significantly different from the estimates for girls. Panel B reveals that differences between rural and urban areas in the effects of the bans were small and statistically insignificant, which is in contrast with the schooling benefits of these bans, documented by Wilson (2022) to be concentrated in urban areas. This suggests that schooling may not be an important mechanism for these child mortality effects.

Table 5: Heterogeneous Effects by Child Gender and Urban Status

Panel A: By child's gender	Infant Mortality			Under-5 Mortality		
	(1) Boys	(2) Girls	(3) Difference	(4) Boys	(5) Girls	(6) Difference
post-ban $\times$ intensity	-0.016*** (0.005)	-0.009*** (0.002)	0.007 (0.006)	-0.028*** (0.009)	-0.025*** (0.004)	0.003 (0.009)
Mean outcome of pre-ban cohorts	0.073	0.060		0.11	0.093	
Observations	103,956	96,619	200,575	103,956	96,619	200,575

Panel B: By rural-urban status	(1) Rural	(2) Urban	(3) Difference	(4) Rural	(5) Urban	(6) Difference
post-ban $\times$ intensity	-0.011*** (0.004)	-0.017*** (0.004)	-0.006 (0.006)	-0.028*** (0.007)	-0.026*** (0.006)	0.001 (0.009)
Mean outcome of pre-ban cohorts	0.078	0.051		0.12	0.073	
Observations	116,501	84,075	200,576	116,501	84,075	200,576

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the ban. All specifications use the weights described in section 3.3 and control for country-cohort fixed effects and subnational region fixed effects.

4.3 Other Outcomes

What are the mechanisms underlying the effects documented above? We know from [Wilson \(2022\)](#) that child marriage bans affected women's decisions related to marriage timing, fertility timing, education (in urban areas only), and employment, as summarized in Appendix Table A4. In theory, all of these are possible mechanisms that could have generated the reductions in child mortality documented above. In Table 6, we use our main regression specification to estimate the effects of child marriage bans on these outcomes in our sample of mothers. We find statistically significant increases in age at first marriage and age at first birth (columns 1 and 2). We do not find any significant effects on educational attainment or current employment (columns 3 and 4). These coefficients are much smaller than those estimated in [Wilson \(2022\)](#), which could be due to treatment effects being smaller in our restricted sample of mothers, or else because of issues related to the selection into motherhood documented in Table 2. We discuss this in more detail in Appendix section C, where we compare results for the full sample of women to those for mothers only.

Overall, our evidence suggests that, although many studies have documented a causal link from parental socioeconomic status (including maternal education and household income) to child health ([Thomas et al., 1991](#); [Currie, 2009](#); [Almond et al., 2018](#)), this does not appear to be an

Table 6: Effects of Child Marriage Bans on Women’s Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
	Age at first marriage	Age at first birth	Years of schooling	Currently employed	Age at first birth < 18	Total fertility
post-ban $\times$ intensity	1.09*** (0.094)	1.03*** (0.071)	0.18 (0.16)	0.016 (0.011)	-0.031*** (0.0071)	-0.85*** (0.059)
Mean outcome of pre- ban cohorts	19.2	20.4	5.41	0.61	0.27	4.13
Observations	191,087	200,581	200,537	200,374	200,581	200,581

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the ban. All specifications use the weights described in section 3.3 and control for country-cohort fixed effects and subnational region fixed effects. Regressions restrict to mothers.

important mechanism in this particular context. This conclusion is also supported by the fact that [Wilson \(2022\)](#) found schooling effects only in urban areas, whereas we found similarly sized mortality effects in both rural and urban areas. In results not reported here but available upon request, we show that changes in husbands’ characteristics do not appear to be an important mechanism either: the bans have no statistically significant effects on current husbands’ education levels or employment status (for women who are currently married or living with their partner).

We therefore focus our attention on the importance of marriage and fertility timing. Age at marriage and first birth could both be important factors because a woman who goes into a marriage or into her first pregnancy with more maturity may have more agency and bargaining power when it comes to securing health investments for herself during pregnancy and for her child immediately after birth. Despite the rich data available in the DHS, we are somewhat limited in our ability to explore specific health-related decisions and investments, which are not available for our entire sample of firstborn children. However, the evidence we do have suggests that child marriage bans increase the likelihood a mother delivers in a private hospital and vaccinates her child (see Appendix Table B1). We present this evidence, along with a description of the main caveats and adjustments for multiple hypothesis testing, in Appendix section B.

In addition, there may also be biological reasons why older maternal age may have improved child survival in our sample. In the medical literature, it has been suggested that the body of a teenager is still not fully developed and not optimal for the development of a successful pregnancy ([Chen et al., 2007](#); [Olausson et al., 1999](#)). We show in column 5 that child marriage bans led to a reduction in the share of mothers giving birth under the age of 18, which is of note because several

studies in developing countries find that children who are born to mothers under 18 have a higher likelihood of dying than those born to older mothers (Neal et al., 2018). That said, while these studies typically control for basic confounders like education and wealth, it is difficult to isolate the biological effect of young maternal age from the effect of all possible omitted variables.

Finally, we also show in column 6 that child marriage bans resulted in a reduction in total fertility (measured at the time of the survey). Because we do not have completed fertility for all mothers in our sample, this reduction could be driven by one (or more) of the following: a delay in first birth, increased birth spacing, or a reduction in total fertility. If child marriage bans lead women to increase birth spacing or reduce total fertility, this would result in more resources being available for the firstborn child in their first years of life, another potential driver of improvements in child mortality.

Given these findings, and to shed light on the heterogeneous mortality effects by country income level, we also examine heterogeneity in the effects on age at marriage, age at first birth, first birth by age 18, and total fertility (the four mother-level variables found to be significantly affected by child marriage bans). Interestingly, Appendix Table A5 does not document substantial heterogeneity in the effects on these variables. Only in panel C, which reports results for the indicator for having a child by age 18, do we find that the low income countries' coefficient is significantly larger in magnitude than that of lower-middle income countries (but not middle income countries). Overall, the patterns of heterogeneity (or lack thereof) in this table do not reflect the same patterns observed for the mortality effects, which were inversely correlated with country income level. This suggests that the heterogeneous mortality effects are unlikely due to larger "first-stage" effects (of child marriage bans on mother's outcomes) in low income countries. Rather, the child health consequences of later marriage as well as later and lower fertility appear to be larger in low-income countries, perhaps because of their lower levels of health, healthcare access, and healthcare quality.

Given that child marriage bans reduced child mortality, a natural follow-up question is whether they also affected intensive-margin health outcomes among surviving children. For certain countries and for children born in the 5 years before the DHS survey, we have measurements of height-for-age, weight-for-age, and weight-for-height z-scores, three commonly used measures of child health. Given that these are only available for surviving children and we know child marriage bans affected survival, regressions on these health outcomes are more complicated to interpret due to potential

compositional changes. Specifically, the children who would have died if it were not for the child marriage bans are likely to be on the lower end of the health distribution, which would result in an underestimate and potential sign reversal of intensive margin health effects. In Appendix Table B2, we repeat our main regression specification using height-for-age, weight-for-age, and weight-for-height z-scores, along with stunting, underweight, and wasting (indicators for z-scores below -2) as our outcome variables, adjusting for multiple hypothesis testing using Benjamini and Hochberg (1995) q-values. We find no evidence of intensive margin health improvements, though compositional changes due to improvements in survival could be masking larger effects.

4.4 Robustness Checks

Our interpretation of the $postban_{kc(r)} \times intensity_r$ coefficient as the causal effect of child marriage bans relies on the assumption that, conditional on all the included fixed effects, the difference between the pre-ban and post-ban cohorts would have remained the same across the treatment intensity distribution in the absence of the bans. A contemporaneous policy that affected child mortality for pre-ban and post-ban cohorts differently, and for which subnational variation in intensity or exposure was correlated with our treatment intensity variable, could violate this assumption. During the early 2000s, numerous developing countries reformed their education systems by either increasing the number of compulsory years of schooling or making education free. Previous studies have shown that these education reforms increase maternal education, which in turn reduces child mortality in developing settings (Grépin and Bharadwaj, 2015; Le et al., 2025; Akresh et al., 2018). We examine whether our main results could be driven by the effects of education reforms by directly controlling for these reforms in our main specification. We search for all education reforms that have been implemented in our study countries.¹⁴ We use the timing of each reform, the schooling grades affected by each reform, and the year of birth of our respondents to generate an indicator for cohort-based exposure to the reform. If these reforms affected everyone in each cohort in each country equally, this would be captured by our cohort-country fixed effects. If, however, there were subnational spatial variation in the intensity of these reforms that was correlated with our treatment intensity variable, then this could be a source of bias. We therefore interact our education reform exposure indicator with our child marriage ban treatment intensity variable and add this to Equation (2) as a control. The results in Panel A of Appendix Table A7

¹⁴Details of these reforms are summarized in Appendix Table A6.

are quantitatively similar to our baseline estimates, suggesting these education reforms were not an important confounder in our main analysis.

In addition to contemporaneous policies, another potential violation could stem from baseline regional characteristics, correlated with our treatment intensity variable, that generate differential trends in child mortality across cohorts. For example, among pre-ban cohorts, areas with high treatment intensity had higher rates of infant and under-5 mortality. This could have resulted in larger improvements in child mortality in these areas due to mean reversion or catch up, even if the bans had not been implemented. To address this potential issue, in panel B of Table A7 we report the results of estimating Equation (2) with additional controls for subnational region-specific characteristics at baseline (infant mortality, under-5 mortality, and women’s years of schooling) interacted with the post-ban dummy.¹⁵ Our results remain statistically significant (and are in fact larger in magnitude than our main estimates), alleviating concerns that differential trends based on these baseline characteristics, rather than treatment intensity, were driving our results above.

Because the DHS does not collect information on the respondents’ place of residence at age 18, we use their current residence to assign the treatment intensity variable. To investigate the extent to which non-classical measurement error in this variable might be biasing our results, we check whether our results hold for those who have never migrated. Restricting our sample to respondents who have never migrated results in a much smaller sample size, though the bulk of this reduction is due to the fact that this question is only asked in 9 countries. Despite the large drop in sample size, however, our estimates are robust to this sample restriction (Panel C, Appendix Table A7).

We test the sensitivity of our estimates to various other sample restrictions. Appendix Figure A2 plots the coefficients and 95% confidence intervals from various specifications that either impose an upper limit, lower limit, or both an upper and lower limit on age at the time of the bans. Panel A displays results restricting the pre-ban cohorts to those aged 35 and under, 40 and under, and 45 and under at the time of the bans; panel B displays results restricting the post-ban cohorts to those 3 and older, 5 and older, and 7 and older at the time of the bans; panel C shows results imposing combinations of the two sets of restrictions. In all panels, coefficients are very similar to those of the main results and remain statistically significant at the 5% level.

All of the results presented above use the weighting procedure described in section 3.3, employed to account for differential selection into motherhood (by age at survey) across the treatment

¹⁵We utilize 12 countries that have a DHS survey prior to the child marriage ban. The five countries that do not have data in the pre-ban period are Albania, Ethiopia, Maldives, the Gambia, and Sierra Leone.

intensity distribution. Another way to address this issue is to directly control for age-by-intensity interactions in our main specification. Because age at survey is highly correlated with age at ban, which is one of our main sources of variation, we prefer the weighting method to this approach. However, we show in Table A8 that our results are not sensitive to this choice. Columns 1 and 2 report results that do not use any weights, columns 3 and 4 report unweighted results with the addition of a linear age-by-intensity interaction, and columns 5 and 6 report unweighted results with the addition of treatment intensity interacted with age fixed effects. Across all specifications, we find statistically significant negative coefficients.

4.5 Selection Concerns

We now explore the plausibility of our explanation for the positive selection coefficient in column 1 of Table 2. We argued above that the positive coefficient was ultimately stemming from the fact that the DHS surveys women at different ages, including future mothers who have not yet had their first child. Therefore, if we were instead able to restrict to women who were old enough to have had their first child or else who have decided not to have children, this would not be an issue. This is unfortunately impossible, given that many countries were surveyed only a few years after the implementation of a child marriage ban, when post-ban women are still relatively young. However, for countries that were surveyed at least 13 years after ban implementation, the youngest post-ban women are 30 years of age at the time of survey. While this is likely not old enough to completely eliminate our proposed selection mechanism, we should see substantial reductions in selection if we restrict to women in these countries aged at least 30.

To explore this, we conduct the following exercise. First, we restrict to the eight countries surveyed at least 13 years after a child marriage ban. Using this sample, we repeat our selection regression using different age cutoffs from 25 to 30. We then repeat our infant and child mortality regressions on this same sample using the same restrictions. In column 1 of Appendix Table A9, we show that the selection coefficient is indeed reduced with each incremental age restriction, from 0.09 (with no age restriction) to 0.03 (restricting to at least 30 years of age). In this last specification, the coefficient is still statistically significant, likely because a 30-year-old cutoff is not old enough to be capturing the first births of all women.¹⁶ Reassuringly, we show in columns 2 and 3 that these restrictions do not affect our main estimates of the effect of child marriage bans

¹⁶We could raise this restriction to at most 34, but this would only leave us with a single country.

on infant and child mortality, which remain similar across panels.

While it is reassuring that (partially) accounting for selection concerns does not substantially change our results, it is important to note that our weighting exercise is not a complete solution to the problem if the “additional” mothers (in high-intensity regions and post-ban cohorts) have systematically different characteristics that affect the survival probabilities of their children. Although we cannot identify these additional mothers, we know they are having their first child at an earlier age than post-ban women in low-intensity regions. If anything, this would suggest lower socioeconomic status (which would be consistent with some suggestive evidence, discussed in Appendix section C, of negative selection based on educational attainment and employment) and therefore less healthy children. If this is the case, our results could be underestimating the magnitude of the mortality effects of child marriage bans.

One way to explore whether selection into motherhood generates compositional changes that could be biasing our results is to use pre-determined characteristics as outcome variables and repeat our main regressions. Given that child marriage bans affected a wide range of outcomes for women, identifying characteristics that could not have been affected by these bans (and which are recorded consistently across countries) is not a trivial task, but religion and sibling mortality are potential options that are included in the majority (though not all) of the countries in our sample. In columns 1 through 7 of Appendix Table A10, we report regressions that use indicators for various religions as outcome variables. All coefficient estimates are small in magnitude and statistically insignificant. In the remaining columns, we explore variables related to sibling death, used by Lu and Vogl (2023) to examine intergenerational persistence in child mortality. We explore two indicator variables, each equal to 1 if the sample mother had any siblings who died before the age of 1 or the age of 5, restricting to deaths that took place before the ban was implemented in their country (to ensure that this variable is indeed pre-determined). Though sibling death is a fairly common event, we find no significant relationship between our main variable of interest and these outcomes. While religion and sibling death only offer a narrow glimpse into the characteristics of these mothers, we do not find any evidence of substantial compositional changes measured using these variables, which is consistent with the stability of our mortality estimates across panels in Table A9.

In theory, maternal mortality is another possible explanation for the positive selection coefficient in Table 2. If child marriage bans reduced maternal mortality, this would increase the relative

share of mothers in post- vs. pre-ban cohorts by a larger amount in high intensity compared to low intensity regions, leading to a positive and significant coefficient in our selection regression. In reality, however, maternal mortality rates in our sample (though high when compared to the rest of the world) are likely to be too low for this to be the case. Among all birth years and countries in both our sample and in the [UN MMEIG et al. \(2024\)](#) series, the median maternal mortality ratio was 604 maternal deaths per 100,000 live births; the highest value was 1,792 (Sierra Leone in 1999).¹⁷ Even if child marriage bans reduced the maternal death ratio to zero, this would not be large enough to generate the 0.08 coefficient in Table 2 or even the 0.03 coefficient in the most restricted specification in Appendix Table A9. Moreover, if child marriage bans did reduce maternal mortality, this would likely lead to an underestimation of the child mortality effects because the additional surviving mothers are likely to be less healthy than average (though we did not detect any compositional changes based on religion or sibling death in Appendix Table A10).

5 Discussion and Conclusion

This paper examines the impacts of increasing the legal minimum age of marriage to 18 on first-born child mortality in 17 developing countries. Using a sample of over 200,000 women drawn from the DHS, we exploit cohort differences in exposure to the ban and regional differences in pre-ban age at marriage to identify causal effects on infant and under-5 mortality. Our main specification documents statistically significant negative effects of child marriage bans on both infant and under-5 mortality, though the infant mortality estimates are not statistically significant in more demanding event-study specifications. Delays in age at marriage and first birth are the most likely channels through which raising the minimum legal age of marriage affected child mortality. Due to child marriage bans, women are going into marriage and motherhood at more mature ages, which could lead to better health-related decisions for their children. For example, using data available only for a restricted sub-sample, we find that child marriage bans increase the likelihood of delivery in a private hospital, as well as the likelihood of a child receiving all basic vaccinations (BCG, DPT, polio, and measles).

The timing of effects deserves some discussion. For under-5 mortality, the event-study analyses in Figures 2 and 3 both report post-ban coefficients that grow larger with longer exposure to the

¹⁷1985 was the earliest year available in the Our World in Data series, and over 90% of our sample was born in 1985 or later.

ban. In Figure 3, coefficient estimates only begin to be statistically significant for girls aged 7 at the time of the ban. What might explain this delayed effect, especially given that Wilson (2022) documents effects on age of marriage and first birth that show up even for those much closer to the age 18 cutoff? One reason is that these “first stage” effects on age of marriage and first birth may need to reach a certain magnitude to generate detectable effects on child mortality – an extreme outcome that is unlikely to be affected for all women whose age of marriage is affected by the bans. Indeed, like our child mortality effects, the effect sizes for these two women’s outcomes also grow larger with greater exposure (Wilson, 2022). There are a number of reasons for this pattern. First, girls who are closer to 18 when a ban is implemented are more likely to already be married or betrothed. Second, it may take time for awareness about a ban to spread across the country, or for enforcement efforts to strengthen. Some countries (like Ethiopia) did not implement an immediate nationwide ban but instead rolled it out across regions over time. In fact, four Ethiopian regions did not implement the ban until 5-9 years after the ban was first implemented in major cities. For these reasons, it might simply take some time for the first stage effects to become large enough to also lead to detectable effects on child mortality.

Findings from our study highlight the importance of evaluating large-scale policies with respect not only to their primary, targeted outcomes but also to potential downstream effects – even when enforcement of such policies is weak, as was the case for some child marriage bans Collin and Talbot (2023). Previous economic literature in developing countries has shown that education reforms and health interventions do not only have positive effects on the directly affected individuals, but also have intergenerational effects on health and mortality (Grépin and Bharadwaj, 2015; Keats, 2018; Akresh et al., 2018; Cornelissen and Dang, 2022; De Neve and Fink, 2018; Walker et al., 2023; Le et al., 2025). Our study contributes to this large body of literature by revealing child marriage bans as yet another government policy that has the potential to improve the health of the next generation.

References

- Akresh, Richard, Daniel Halim, and Marieke Kleemans. 2018. “Long-Term and Intergenerational Effects of Education: Evidence from School Construction in Indonesia.” *The Economic Journal* 133 (650): 582–612. [10.3386/w25265](#).
- Almond, Douglas, Janet Currie, and Valentina Duque. 2018. “Childhood Circumstances and Adult Outcomes: Act II.” *Journal of Economic Literature* 56 (4): 1360–1446. [10.1257/jel.20171164](#).
- Arthur, Megan, Alison Earle, Amy Raub et al. 2018. “Child Marriage Laws around the World: Minimum Marriage Age, Legal Exceptions, and Gender Disparities.” *Journal of Women, Politics & Policy* 39 (1): 51–74. [10.1080/1554477X.2017.1375786](#).
- Asadullah, M Niaz, and Zaki Wahhaj. 2019. “Early Marriage, Social Networks and the Transmission of Norms.” *Economica* 86 (344): 801–831. [10.1111/ecca.12291](#).
- Baird, Sarah, Jed Friedman, and Norbert Schady. 2011. “Aggregate Income Shocks and Infant Mortality in the Developing World.” *Review of Economics and Statistics* 93 (3): 847–856. [10.1162/REST_a_00084](#).
- Barofsky, Jeremy, Tobenna D Anekwe, and Claire Chase. 2015. “Malaria Eradication and Economic Outcomes in Sub-Saharan Africa: Evidence from Uganda.” *Journal of Health Economics* 44 118–136. [10.1016/j.jhealeco.2015.08.002](#).
- Batyra, Ewa, and Luca Maria Pesando. 2021. “Trends in Child Marriage and New Evidence on the Selective Impact of Changes in Age-at-Marriage Laws on Early Marriage.” *SSM - Population Health* 14 100811. [10.1016/j.ssmph.2021.100811](#).
- Bellés-Obrero, Cristina, and María Lombardi. 2023. “Will You Marry Me, Later?” *Journal of Human Resources* 58 (1): 221–259. [10.3368/jhr.58.3.1219-10621R2](#).
- Benjamini, Yoav, and Yosef Hochberg. 1995. “Controlling the false discovery rate: a practical and powerful approach to multiple testing.” *Journal of the Royal statistical society: series B (Methodological)* 57 (1): 289–300.

- Bergstrom, Katy, and Berk Özler.** 2023. “Improving the Well-Being of Adolescent Girls in Developing Countries.” *The World Bank Research Observer* 38 (2): 179–212. [10.1093/wbro/lkac007](#).
- Bharadwaj, Prashant.** 2015. “Impact of changes in marriage law: Implications for fertility and school enrollment.” *Journal of Human Resources* 50 (3): 614–654. [10.3368/jhr.50.3.614](#).
- Bhuwania, Pragya, Kate Huh, and Jody Heymann.** 2023. “Impact of Tuition-Free Education Policy on Child Marriage and Early Childbearing: Does Secondary Matter More?” *Population and Development Review* 49 (1): 43–70. [10.1111/padr.12538](#).
- Bleakley, H.** 2007. “Disease and Development: Evidence from Hookworm Eradication in the American South.” *The Quarterly Journal of Economics* 122 (1): 73–117. [10.1162/qjec.121.1.73](#).
- Buchmann, Nina, Erica Field, Rachel Glennerster, Shahana Nazneen, and Xiao Yu Wang.** 2023. “A Signal to End Child Marriage: Theory and Experimental Evidence from Bangladesh.” *American Economic Review* 113 (10): 2645–2688. [10.1257/aer.20220720](#).
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro HC Sant’Anna.** 2024. “Event-Studies with a Continuous Treatment.” [10.3386/w32118](#), NBER Working Paper No. w32118.
- Callaway, Brantly, and Pedro HC Sant’Anna.** 2021. “Difference-in-differences with multiple time periods.” *Journal of econometrics* 225 (2): 200–230. [10.1016/j.jeconom.2020.12.001](#).
- Chari, A V, Rachel Heath, Annemie Maertens, and Freeha Fatima.** 2017. “The Causal Effect of Maternal Age at Marriage on Child Wellbeing: Evidence from India.” *Journal of Development Economics* 127 42–55. [10.1016/j.jdeveco.2017.02.002](#).
- Chen, X-K, S W Wen, N Fleming, K Demissie, G G Rhoads, and M Walker.** 2007. “Teenage Pregnancy and Adverse Birth Outcomes: A Large Population Based Retrospective Cohort Study.” *International Journal of Epidemiology* 36 (2): 368–373. [10.1093/ije/dyl284](#).
- Chou, Shin-Yi, Jin-Tan Liu, Michael Grossman, and Ted Joyce.** 2010. “Parental Education and Child Health: Evidence from a Natural Experiment in Taiwan.” *American Economic Journal: Applied Economics* 2 (1): 33–61. [10.1257/app.2.1.33](#).

- Chow, Vinci, and Eva Vivalt.** 2022. “Challenges in Changing Social Norms: Evidence from Interventions Targeting Child Marriage in Ethiopia.” *Journal of African Economies* 31 (3): 183–210. [10.1093/jae/ejab010](#).
- Cohen, Isabelle, Maryam Abubakar, and Daniel Perlman.** 2023. “Pathways to Choice: A Bundled Intervention against Child Marriage.” [10.26085/C31C71](#), CEGA Working Paper Series, No. WPS-230.
- Collin, Matthew, and Theodore Talbot.** 2023. “Are Age-of-Marriage Laws Enforced? Evidence from Developing Countries.” *Journal of Development Economics* 160 102950. [10.1016/j.jdeveco.2022.102950](#).
- Cornelissen, Thomas, and Thang Dang.** 2022. “The Multigenerational Impacts of Educational Expansion: Evidence from Vietnam.” *Labour Economics* 78 102243. [10.1016/j.labeco.2022.102243](#).
- Corno, Lucia, Nicole Hildebrandt, and Alessandra Voena.** 2020. “Age of marriage, weather shocks, and the direction of marriage payments.” *Econometrica* 88 (3): 879–915.
- Corno, Lucia, and Alessandra Voena.** 2023. “Child marriage as informal insurance: Empirical evidence and policy simulations.” *Journal of Development Economics* 162 103047.
- Currie, Janet.** 2009. “Healthy, Wealthy, and Wise: Socioeconomic Status, Poor Health in Childhood, and Human Capital Development.” *Journal of Economic Literature* 47 (1): 87–122. [10.1257/jel.47.1.87](#).
- Cutler, David, Winnie Fung, Michael Kremer, Monica Singhal, and Tom Vogl.** 2010. “Early-Life Malaria Exposure and Adult Outcomes: Evidence from Malaria Eradication in India.” *American Economic Journal: Applied Economics* 2 (2): 72–94. [10.1257/app.2.2.72](#).
- Dahl, Gordon B.** 2010. “Early teen marriage and future poverty.” *Demography* 47 (3): 689–718. [10.1353/dem.0.0120](#).
- De Neve, Jan-Walter, and Günther Fink.** 2018. “Children’s Education and Parental Old Age Survival – Quasi-Experimental Evidence on the Intergenerational Effects of Human Capital Investment.” *Journal of Health Economics* 58 76–89. [10.1016/j.jhealeco.2018.01.008](#).

- Dursun, Bahadır, Resul Cesur, and Inas Rashad Kelly.** 2017. “The Value of Mandating Maternal Education in a Developing Country.” [10.3386/w23492](#), NBER Working Paper, No. w23492.
- Field, Erica, and Attila Ambrus.** 2008. “Early Marriage, Age of Menarche, and Female Schooling Attainment in Bangladesh.” *Journal of Political Economy* 116 (5): 881–930. [10.1086/593333](#).
- Garcia-Hombrados, Jorge.** 2022. “Child Marriage and Infant Mortality: Causal Evidence from Ethiopia.” *Journal of Population Economics* 35 (3): 1163–1223. [10.1007/s00148-021-00873-y](#).
- Giacobino, Hélène, Elise Huillery, Bastien Michel, and Mathilde Sage.** 2024. “Schoolgirls, not brides: Education as a shield against child marriage.” *American Economic Journal: Applied Economics* 16 (4): 109–143.
- Grépin, Karen A, and Prashant Bharadwaj.** 2015. “Maternal Education and Child Mortality in Zimbabwe.” *Journal of Health Economics* 44 97–117. [10.1016/j.jhealeco.2015.08.003](#).
- Huang, Jian, Wim Groot, John G Sessions, and Yinyen Tseng.** 2019. “Age of Menarche, Adolescent Sexual Intercourse and Schooling Attainment of Women.” *Oxford Bulletin of Economics and Statistics* 81 (4): 717–743. [10.1111/obes.12284](#).
- ICF.** 2018. “Demographic and Health Surveys.” <https://dhsprogram.com/>, [accessed 01 December 2022].
- Kammerlander, Andreas, and Günther G Schulze.** 2023. “Local Economic Growth and Infant Mortality.” *Journal of Health Economics* 87 102699. [10.1016/j.jhealeco.2022.102699](#).
- Keats, Anthony.** 2018. “Women’s Schooling, Fertility, and Child Health Outcomes: Evidence from Uganda’s Free Primary Education Program.” *Journal of Development Economics* 135 142–159. [10.1016/j.jdeveco.2018.07.002](#).
- Kuecken, Maria, Josselin Thuilliez, and Marie-Anne Valfort.** 2021. “Disease and Human Capital Accumulation: Evidence from the Roll Back Malaria Partnership in Africa.” *The Economic Journal* 131 (637): 2171–2202. [10.1093/ej/ueaa134](#).
- Le, Duc Dung, Thac Minh Nguyen, and Yoko Ibuka.** 2025. “Intergenerational effects of education on child mortality: Evidence from the compulsory primary schooling law in Vietnam.” *World Development* 185. [10.1016/j.worlddev.2024.106810](#).

- Lu, Frances, and Tom Vogl.** 2023. “Intergenerational Persistence in Child Mortality.” *American Economic Review: Insights* 5 (1): 93–109. [10.1257/aeri.20210604](https://doi.org/10.1257/aeri.20210604).
- Lucas, Adrienne M.** 2010. “Malaria Eradication and Educational Attainment: Evidence from Paraguay and Sri Lanka.” *American Economic Journal: Applied Economics* 2 (2): 46–71. [10.1257/app.2.2.46](https://doi.org/10.1257/app.2.2.46).
- McGavock, Tamara.** 2021. “Here Waits the Bride? The Effect of Ethiopia’s Child Marriage Law.” *Journal of Development Economics* 149 102580. [10.1016/j.jdeveco.2020.102580](https://doi.org/10.1016/j.jdeveco.2020.102580).
- Neal, Sarah, Andrew Amos Channon, and Jesman Chintsanya.** 2018. “The Impact of Young Maternal Age at Birth on Neonatal Mortality: Evidence from 45 Low and Middle Income Countries.” *PLOS ONE* 13 (5): e0195731. [10.1371/journal.pone.0195731](https://doi.org/10.1371/journal.pone.0195731).
- Olausson, Petra Otterblad, Sven Cnattingius, and Bengt Haglund.** 1999. “Teenage Pregnancies and Risk of Late Fetal Death and Infant Mortality.” *BJOG: An International Journal of Obstetrics and Gynaecology* 106 (2): 116–121. [10.1111/j.1471-0528.1999.tb08210.x](https://doi.org/10.1111/j.1471-0528.1999.tb08210.x).
- Psaki, Stephanie R, Andrea J Melnikas, Eashita Haque, Grace Saul, Christina Misunas, Sangram K Patel, Thoai Ngo, and Sajeda Amin.** 2021. “What are the drivers of child marriage? A conceptual framework to guide policies and programs.” *Journal of Adolescent Health* 69 (6): S13–S22.
- Rokicki, Slawa.** 2021. “Impact of Family Law Reform on Adolescent Reproductive Health in Ethiopia: A Quasi-Experimental Study.” *World Development* 144 105484. [10.1016/j.worlddev.2021.105484](https://doi.org/10.1016/j.worlddev.2021.105484).
- Roth, Jonathan, Pedro HC Sant’Anna, Alyssa Bilinski, and John Poe.** 2023. “What’s trending in difference-in-differences? A synthesis of the recent econometrics literature.” *Journal of Econometrics* 235 (2): 2218–2244. [10.1016/j.jeconom.2023.03.008](https://doi.org/10.1016/j.jeconom.2023.03.008).
- Sekhri, Sheetal, and Sisir Debnath.** 2014. “Intergenerational Consequences of Early Age Marriages of Girls: Effect on Children’s Human Capital.” *Journal of Development Studies* 50 (12): 1351–1368. [10.1080/00220388.2014.915657](https://doi.org/10.1080/00220388.2014.915657).
- Sunder, Naveen.** 2019. “Marriage Age, Social Status, and Intergenerational Effects in Uganda.” *Demography* 56 (6): 2123–2146. [10.1007/s13524-019-00829-8](https://doi.org/10.1007/s13524-019-00829-8).

- Tauseef, Salauddin, and Farha Deba Sufian.** 2024. “The Causal Effect of Early Marriage on Women’s Bargaining Power: Evidence from Bangladesh.” *The World Bank Economic Review* lhad046.
- Thomas, Duncan, John Strauss, and Maria-Helena Henriques.** 1991. “How does mother’s education affect child height?” *Journal of human resources* 26 (2): 183–211. [10.2307/145920](#).
- UN MMEIG, WHO, United Nations, and Gapminder.** 2024. “Maternal mortality ratio [dataset].” <https://ourworldindata.org/grapher/maternal-mortality>, Data sources: UN MMEIG (2023); WHO Mortality Database (2024); UN World Population Prospects (2024); Gapminder (2010). Processed and curated by Our World in Data. Original sources include: “Trends in maternal mortality 2020” (UN MMEIG); “WHO Mortality Database”; “World Population Prospects” (UN); “Maternal mortality ratio V1” (Gapminder).
- UNFPA-UNICEF.** 2019. “Investing in Knowledge for Ending Child Marriage: Publications Catalogue 2018-2019.” <https://www.unicef.org/bulgaria/media/6111/file/BGR-Investing-in-knowledge-for-ending-child-marriage.pdf>, [accessed 10 March 2023].
- UNICEF.** 2017. “Ending Child Marriage - Progress and prospects.” June, <https://www.refworld.org/reference/themreport/unicef/2017/en/117022>, [accessed 16 April 2024].
- United Nations.** 1995. “Beijing Declaration and Platform for Action. 16th Plenary Meeting.” <https://www.un.org/womenwatch/daw/beijing/pdf/BDPfA%20E.pdf>, [accessed 8 March 2023].
- United Nations Inter-agency Group for Child Mortality Estimation.** 2018. “Levels & Trends in Child Mortality: Report 2018, Estimates Developed by the United Nations Inter-Agency Group for Child Mortality Estimation.” <https://www.unicef.org/reports/levels-and-trends-child-mortality-report-2018>, [accessed 8 March 2023].
- Walker, Michael W, Alice H Huang, Suleiman Asman et al.** 2023. “Intergenerational child mortality impacts of deworming: Experimental evidence from two decades of the Kenya Life Panel Survey.” [10.3386/w31162](#), NBER Working Paper, No. w31162.
- Wang, Chunbei, and Le Wang.** 2017. “Knot yet: minimum marriage age law, marriage delay, and earnings.” *Journal of Population Economics* 30 771–804. [10.1007/s00148-017-0632-5](#).

- Wilson, Nicholas.** 2022. “Child Marriage Bans and Female Schooling and Labor Market Outcomes: Evidence from Natural Experiments in 17 Low- and Middle-Income Countries.” *American Economic Journal: Economic Policy* 14 (3): 449–477. [10.1257/pol.20200008](https://doi.org/10.1257/pol.20200008).
- Wodon, Quentin, Chata Male, Ada Nayihouba, Adenike Onagoruwa, Abouduhyme Savadogo, Ali Yedan, and Jeff Edmeades.** 2017. “Economic Impacts of Child Marriage: Global Synthesis Report.” <https://documents1.worldbank.org/curated/en/530891498511398503/pdf/116829-WP-P151842-PUBLIC-EICM-Global-Conference-Edition-June-27.pdf>, [accessed 20 February 2023].
- World Policy Center.** 2022. “Marriage.” <https://www.worldpolicycenter.org/topics/childmarriage/policies>, [accessed 01 December 2022].

A Appendix

Table A1: Child Marriage Bans and DHS Survey Years

Country	Ban year	Survey year	Years between ban and survey
Albania	2003	2017-18	14-15
Benin	2004	2017-18	13-14
D.R.C.	2009	2013-14	4-5
Egypt	2008	2014	6
Ethiopia	2000	2016	16
Guinea	2008	2018	10
Jordan	2001	2017-18	16-17
Kazakhstan	1998	1999	1
Liberia	2012	2014	2
Madagascar	2007	2008-09	1-2
Maldives	2001	2016-17	15-16
Namibia	1996	2013	17
Nepal	2003	2016	13
Nigeria	2003	2018	15
Peru	1999	2000	1
Sierra Leone	2007	2013	6
The Gambia	2005	2013-14	8-9
Togo	2007	2013	6

Data sources: The World Policy Analysis Center and the Demographic and Health Surveys. D.R.C denotes Democratic Republic of the Congo. Kazakhstan is not included in the main analysis.

Table A2: Summary Statistics for Potential Mechanisms

	(1)	(2)
	Unweighted	Weighted
Years of schooling	5.65 (5.30) 200,537	5.61 (5.29) 200,537
Currently employed	0.58 (0.49) 200,374	0.58 (0.49) 200,374
Age at first marriage	18.9 (4.52) 191,088	18.8 (4.47) 191,088
Age at first birth	20.1 (4.22) 200,581	20.0 (4.15) 200,581
Age at first birth < 18	0.29 (0.45) 200,581	0.29 (0.45) 200,581
Total fertility	3.61 (2.30) 200,581	3.70 (2.38) 200,581

Notes: Means, standard deviations (in parentheses), and observation counts are reported. Sample restricts to women who have ever given birth and who were born before the child marriage ban was implemented in their country. Column 2 weights the sample using the methods described in section 3.3.

Table A3: Availability of Health-Seeking and Child Health Outcome Variables

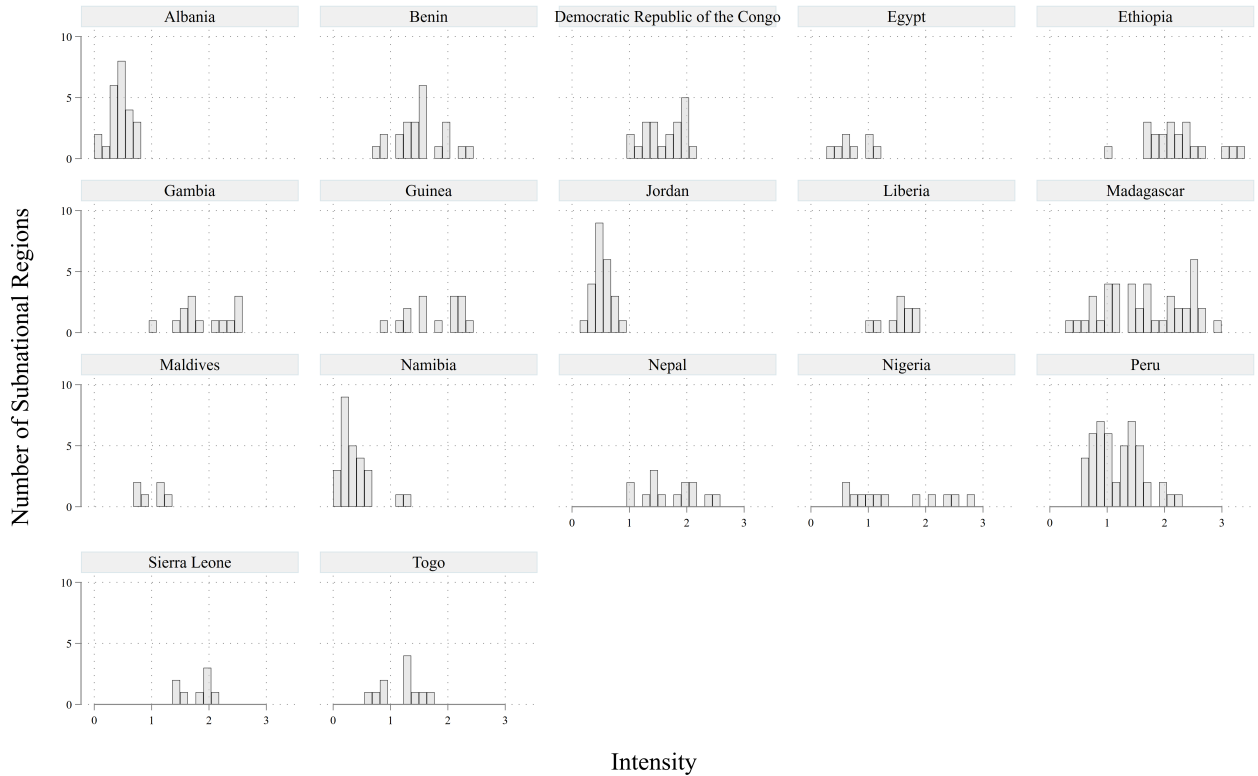
Variable	Availability
The number of antenatal care visits	All countries: The most recent child born in the 5 years before the survey
Receiving antenatal care at least four times	All countries: The most recent child born in the 5 years before the survey
Antenatal care quality	All countries: The most recent child born in the 5 years before the survey
Birth delivery: Home	All countries: The most recent child born in the 5 years before the survey
Birth delivery: Gov. Hospital	All countries: The most recent child born in the 5 years before the survey
Birth delivery: Gov. Other	All countries: The most recent child born in the 5 years before the survey
Birth delivery: Pvt. Hospital	All countries: The most recent child born in the 5 years before the survey
Birth delivery: Pvt. Other	All countries: The most recent child born in the 5 years before the survey
All vaccinations	All countries (except Albania): Surviving children born in the 3 or 5 years before the survey
Vaccinated: BCG	All countries: Surviving children born in the 3 (Benin, Ethiopia, Guinea, Jordan, Maldives, Nepal, and Nigeria) or 5 years (D.R. Congo, Egypt, Gambia, Liberia, Madagascar, Namibia, Peru, Sierra Leone, and Togo) before the survey
Vaccinated: DPT	All countries: Surviving children born in the 3 (Benin, Ethiopia, Guinea, Jordan, Maldives, Nepal, and Nigeria) or 5 years (D.R. Congo, Egypt, Gambia, Liberia, Madagascar, Namibia, Peru, Sierra Leone, and Togo) before the survey
Vaccinated: Polio	All countries: Surviving children born in the 3 (Benin, Ethiopia, Guinea, Jordan, Maldives, Nepal, and Nigeria) or 5 years (D.C.R. Congo, Egypt, Gambia, Liberia, Madagascar, Namibia, Peru, Sierra Leone, and Togo) before the survey
Vaccinated: Measles	All countries: Surviving children born in the 3 (Benin, Ethiopia, Guinea, Jordan, Maldives, Nepal, and Nigeria) or 5 years (D.C.R. Congo, Egypt, Gambia, Liberia, Madagascar, Namibia, Peru, Sierra Leone, and Togo) before the survey
Height-for-Age	All countries (except Jordan): Surviving children born in the 5 years before the survey
Weight-for-Age	All countries (except Jordan and Madagascar): Surviving children born in the 5 years before the survey
Weight-for-Height	All countries (except Jordan and Madagascar): Surviving children born in the 5 years before the survey
Stunting	All countries (except Jordan): Surviving children born in the 5 years before the survey
Underweight	All countries (except Jordan and Madagascar): Surviving children born in the 5 years before the survey
Wasting	All countries (except Jordan and Madagascar): Surviving children born in the 5 years before the survey

Table A4: Summary of Wilson (2022) Main Results

Outcome variable	Sample		
	Full Sample	Urban	Rural
Married before age 18	Not significant	Negative, significant	Not significant
Age at marriage	Positive, significant	Positive, significant	Positive, significant
Age at first birth	Positive, significant	Positive, significant	Positive, significant
Years of schooling	Not significant	Positive, significant	Negative, significant
Employed	Positive, significant	Positive, significant	Positive, significant

Notes: Table indicates statistical significance (at the 5% level) of the coefficient of interest and reports the sign only for significant coefficients.

Figure A1: Within-country Distributions of Treatment Intensity Variable



Graphs by country

Notes: Treatment intensity is calculated as described in Equation (1).

Table A5: Heterogeneous Effects on Mothers' Characteristics by Country Income Classification

	(1)	(2)	(3)	(4)
	Low Income	Lower-Middle Income	Middle Income	Difference
Panel A: Age at first marriage				
post-ban $\times$ intensity	1.147*** (0.173)	1.046*** (0.103)	1.849*** (0.402)	
Low-Mid vs. Low				-0.101 (0.200)
Middle vs. Low				0.702 (0.436)
Mean outcome of pre-ban cohorts	18.2	19.6	20.6	
Observations	74,493	69,449	47,145	191,087
Panel B: Age at first birth				
post-ban $\times$ intensity	0.908*** (0.163)	1.066*** (0.082)	0.947*** (0.266)	
Low-Mid vs. Low				0.157 (0.182)
Middle vs. Low				0.039 (0.311)
Mean outcome of pre-ban cohorts	19.4	21.1	21.5	
Observations	78,927	70,744	50,910	200,581
Panel C: Age at first birth ≤ 18				
post-ban $\times$ intensity	-0.062*** (0.019)	-0.022*** (0.007)	-0.041 (0.033)	
Low-Mid vs. Low				0.040** (0.020)
Middle vs. Low				0.020 (0.038)
Mean outcome of pre-ban cohorts	0.35	0.22	0.17	
Observations	78,927	70,744	50,910	200,581
Panel D: Total fertility				
post-ban $\times$ intensity	-0.777*** (0.201)	-0.863*** (0.055)	-0.839*** (0.160)	
Low-Mid vs. Low				-0.086 (0.208)
Middle vs. Low				-0.061 (0.257)
Mean outcome of pre-ban cohorts	4.35	4.28	3.55	
Observations	78,927	70,744	50,910	200,581

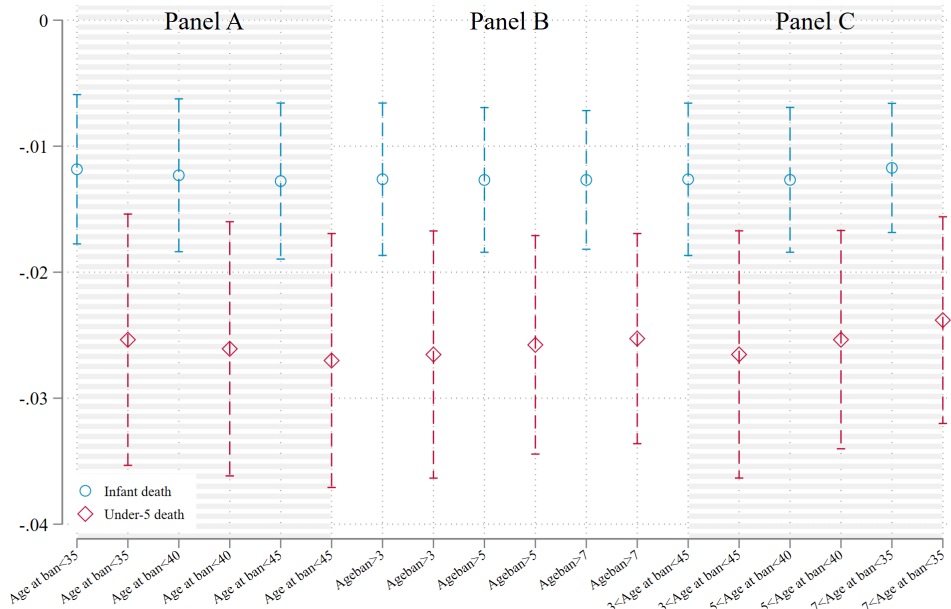
Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications use the weights described in section 3.3 and control for country-cohort fixed effects and subnational region fixed effects. Country income levels are taken from World Bank income classifications. Regressions restrict to mothers.

Table A6: Education Reforms in Study Countries

Country	Type of reforms	Year of reforms	Primary/ secondary school starting age	Cohorts exposed to reforms
Albania	Increase compulsory school from 8 to 9	2012	6/11	1998
D.R.C.	Free primary and secondary schools	2015	6/12	2009
Egypt	Increase compulsory school from 9 to 12	2015	6/12	2002
Liberia	Free primary and secondary schools	2003/2011	6/12	1997/1999
Nigeria	Free primary and secondary schools	2004	6/12	1998
Peru	Increase compulsory school from 6 to 8	2012	6/13	2000
Sierra Leone	Free primary and secondary schools	2000/2004	6/12	1994
The Gambia	Free primary and secondary schools	2004	7/13	1997

Notes: Education reform information is taken from previous literature (Bhuwania et al., 2023; Wilson, 2022). D.R.C denotes Democratic Republic of the Congo.

Figure A2: Coefficient estimates for alternative samples



Notes: Standard errors clustered at region level. Figure reports coefficients (and 95% confidence intervals) on the postban-intensity interaction term in Equation (2), using the weighting procedure described in section 3.3, for various alternative samples based on age at the time of the ban.

Table A7: Robustness tests: Education Reforms, Region-specific Trends, and Migration

	(1)	(2)
	Infant Mortality	Under-5 Mortality
Panel A: controlling for education reforms		
post-ban $\times$ intensity	-0.017*** (0.003)	-0.026*** (0.004)
Mean outcome of pre-ban cohorts	0.067	0.100
Observations	200,581	200,581
Panel B: allowing for differential trends by baseline characteristics		
	(1)	(2)
	Infant Mortality	Under-5 Mortality
post-ban $\times$ intensity	-0.016*** (0.004)	-0.029*** (0.008)
Mean outcome of pre-ban cohorts	0.063	0.095
Observations	158,420	158,420
Panel C: restricting to non-movers		
	(1)	(2)
	Infant Mortality	Under-5 Mortality
post-ban $\times$ intensity	-0.015*** (0.005)	-0.029*** (0.006)
Mean outcome of pre-ban cohorts	0.061	0.094
Observations	66,475	66,475

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications use the weights described in section 3.3 and control for country-cohort fixed effects and subnational region fixed effects. Panel A additionally controls for an interaction between an education reform exposure indicator and the treatment intensity variable. Using only 12 countries that have data available in the pre-ban period, Panel B additionally controls for the post-ban dummy interacted with subnational region-specific characteristics at baseline (infant mortality, under-5 mortality, and years of schooling). Panel C restricts the analysis sample to those who are still living in their place of birth (in the 9 where this information was recorded).

Table A8: Effects of Child Marriage Bans on First-born Child Mortality: Unweighted estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	Infant Mortality	Under-5 Mortality	Infant Mortality	Under-5 Mortality	Infant Mortality	Under-5 Mortality
post-ban $\times$ intensity	-0.012*** (0.003)	-0.026*** (0.005)	-0.008** (0.003)	-0.010** (0.004)	-0.010*** (0.004)	-0.014*** (0.005)
Added Survey Age Controls	None	None	Linear Age $\times$ Intensity	Linear Age $\times$ Intensity	Age Dummies $\times$ Intensity	Age Dummies $\times$ Intensity
Mean outcome of pre-ban cohorts	0.067	0.100	0.067	0.100	0.067	0.100
Observations	200,581	200,581	200,581	200,581	200,581	200,581

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications are unweighted and control for country-cohort fixed effects and subnational region fixed effects.

Table A9: Selection and Mortality Effects, Restricting by Mothers' Age at Survey

	(1)	(2)	(3)
	In Mother Sample	Infant Mortality	Under-5 Mortality
Panel A: All ages			
post-ban $\times$ intensity	0.086*** (0.013)	-0.011*** (0.003)	-0.024*** (0.006)
Mean outcome of pre-ban cohorts	0.95	0.058	0.088
Observations	128,024	93,228	93,228
Panel B: Ages 25 and older			
post-ban $\times$ intensity	0.068*** (0.006)	-0.007*** (0.003)	-0.017*** (0.005)
Mean outcome of pre-ban cohorts	0.95	0.058	0.088
Observations	84,935	78,072	78,072
Panel C: Ages 26 and older			
post-ban $\times$ intensity	0.058*** (0.006)	-0.009*** (0.003)	-0.017*** (0.006)
Mean outcome of pre-ban cohorts	0.95	0.058	0.088
Observations	79,102	73,327	73,327
Panel D: Ages 27 and older			
post-ban $\times$ intensity	0.048*** (0.005)	-0.011*** (0.004)	-0.017*** (0.006)
Mean outcome of pre-ban cohorts	0.95	0.057	0.088
Observations	75,051	70,077	70,077
Panel E: Ages 28 and older			
post-ban $\times$ intensity	0.043*** (0.005)	-0.011*** (0.004)	-0.017*** (0.006)
Mean outcome of pre-ban cohorts	0.95	0.057	0.087
Observations	70,870	66,563	66,563
Panel F: Ages 29 and older			
post-ban $\times$ intensity	0.036*** (0.005)	-0.009* (0.005)	-0.014** (0.007)
Mean outcome of pre-ban cohorts	0.95	0.057	0.087
Observations	66,319	62,583	62,583
Panel G: Ages 30 and older			
post-ban $\times$ intensity	0.032*** (0.006)	-0.014*** (0.003)	-0.020*** (0.006)
Mean outcome of pre-ban cohorts	0.95	0.057	0.087
Observations	62,786	59,446	59,446

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions restrict to the eight countries surveyed at least 13 years after the implementation of a child marriage ban. *post-ban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications control for country-cohort fixed effects and subnational region fixed effects.

Table A10: Exploring Compositional Changes

	(1) Muslim	(2) Catholic	(3) Protestant	(4) Other Christian	(5) Hindu	(6) Others	(7) No religion	(8) Sibling Infant Death	(9) Sibling Under-5 Death
post-ban $\times$ intensity	-0.004 (0.007)	0.004 (0.003)	-0.002 (0.002)	0.000 (0.008)	0.001 (0.001)	0.002 (0.003)	-0.000 (0.001)	-0.003 (0.003)	-0.007 (0.004)
Mean out- come of pre-ban cohorts	0.45	0.17	0.087	0.12	0.041	0.093	0.034	0.18	0.27
Observations	163,207	163,207	163,207	163,207	163,207	163,207	163,207	142,848	142,848

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications use the weights described in section 3.3 and control for country-cohort fixed effects and subnational region fixed effects. Columns 1-7 exclude Jordan, the Maldives, and Peru, which do not have information about religion. Columns 8 and 9 exclude Albania, Egypt, Guinea, Jordan, and the Maldives, which do not have information on sibling deaths.

B Other Outcomes

In this section, we explore the effects of child marriage bans on variables relating to child health investments and child health outcomes. Unfortunately, all of the variables explored here are only available for a subset of our original sample, as summarized in Appendix Table A3. Therefore, effects we estimate here may not be representative of the first-born children in our main analysis. In addition, because some of these variables are only available for children who survive until the survey is fielded, coefficient estimates may also be affected by compositional changes driven by the bans' effects on child mortality.

With these caveats in mind, we repeat our main regressions on various pre-natal, delivery, and post-natal outcomes, which represent potential mechanisms through which bans might improve child survival. We use the same weighting procedure that equalizes the age distribution across regions, but because different regressions use markedly different sub-samples, different weights (which equalize the age distributions across regions) are calculated for each sub-sample. We report results in panel A of Appendix Table B1, including a row of Benjamini and Hochberg (1995) q-values that adjust for multiple hypothesis testing. In panels B and C, for outcomes which are available even for children who have died, we repeat our main mortality regressions using the sub-sample used in column 1, to verify that our estimated mortality effects are not substantially different in these sub-samples.

We find no significant effects of child marriage bans on antenatal care, either the number of visits or quality of care (measured as the sum of three indicator variables for having had a urine sample taken, blood sample taken, and received iron pills during pregnancy). However, we do find that child marriage bans change the location of delivery, increasing the likelihood of delivery at a private hospital (column 7). Finally, we find that child marriage bans increase the likelihood of a child receiving all shots for the BCG, DPT, polio, and measles vaccines (driven primarily by increases in the measles vaccine). Unlike the delivery outcomes, which are available for both surviving and non-surviving children, vaccination information is only available for surviving children, which means potential compositional changes complicate the interpretation of these estimates.

Table B1: Effects of Child Marriage Bans on Health-Seeking Behavior

	(1) Number of ANC visits	(2) Attended 4+ ANC visits	(3) Antenatal Care Quality	(4) Delivery: Home	(5) Delivery: Gov. Hospital	(6) Delivery: Gov. Other	(7) Delivery: Pvt. Hospital	(8) Delivery: Pvt. Other	(9) All Vaccinations	(10) Vaccinated: BCG	(11) Vaccinated: DPT	(12) Vaccinated: Polio	(13) Vaccinated: Measles
Panel A: Health-Seeking Outcomes													
post-ban $\times$ intensity	-0.125 (0.263)	-0.001 (0.018)	0.018 (0.056)	-0.018 (0.022)	0.014 (0.016)	-0.028 (0.019)	0.033** (0.014)	-0.006 (0.008)	0.102*** (0.017)	-0.002 (0.011)	0.024 (0.019)	0.034* (0.019)	0.079*** (0.017)
BH q-value	0.827	0.944	0.877	0.622	0.622	0.388	0.078	0.622	0.000	0.944	0.414	0.270	0.000
Mean outcome of pre-ban cohorts	7.2	.79	1.8	.24	.28	.24	.13	.11	.51	.93	.76	.72	.67
Observations	30,011	30,011	30,011	43,011	43,011	43,011	43,011	43,011	33,441	34,206	33,616	33,991	33,989
Panel B: Infant Mortality Results for Sub-Sample in Panel A													
post-ban $\times$ intensity	-0.018** (0.007)	-0.018** (0.007)	-0.018** (0.007)	-0.013 (0.008)	-0.013 (0.008)	-0.013 (0.008)	-0.013 (0.008)	-0.013 (0.008)					
Mean outcome of pre-ban cohorts	0.028	0.028	0.028	0.040	0.040	0.040	0.040	0.040					
Observations	30,011	30,011	30,011	43,011	43,011	43,011	43,011	43,011					
Panel C: Child Mortality Results Results for Sub-Sample in Panel A													
post-ban $\times$ intensity	-0.027*** (0.009)	-0.027*** (0.009)	-0.027*** (0.009)	-0.019* (0.011)	-0.019* (0.011)	-0.019* (0.011)	-0.019* (0.011)	-0.019* (0.011)					
Mean outcome of pre-ban cohorts	0.033	0.033	0.033	0.050	0.050	0.050	0.050	0.050					
Observations	30,011	30,011	30,011	43,011	43,011	43,011	43,011	43,011					

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the of the ban. All specifications use the weights described in section 3.3, calculated separately for each regression, and control for country-cohort fixed effects and subnational region fixed effects. Panel A reports regression results on the outcomes listed as column titles. “BH q-value” refers to the [Benjamini and Hochberg \(1995\)](#) q-value that adjusts for multiple hypothesis testing. Panels B and C report regression results on infant mortality and child mortality, respectively, restricting to the sub-sample used in the Panel A regressions. Appendix Table [A3](#) provides information about the availability of these variable by country and child age.

In Table B2, we explore effects of the bans on various measures of child health: height-for-age, weight-for-age, and weight-for-height z-scores, along with the corresponding indicators for z-scores below -2: stunting, underweight, and wasting. Adjusting for multiple hypothesis testing, we find no evidence of the bans significantly affecting these intensive margin measures of child health. However, it is important to note that these coefficients could be underestimating or even have the opposite sign of the true effects, due to the possibility that the reductions in child mortality caused by child marriage bans led to the survival of the least healthy babies. For example, the positive coefficient in column 6 (statistically significant only when we do not adjust for multiple hypothesis testing) could be due to the fact that child marriage bans improved child survival among babies with the lowest weight-for-height z-scores.

Table B2: Effects of Child marriage Bans on Child Health Outcomes

	(1) Height-for- Age	(2) Weight-for- Age	(3) Weight-for- Height	(4) Stunting	(5) Underweight	(6) Wasting
post-ban $\times$ intensity	0.098 (0.119)	0.107 (0.075)	0.019 (0.077)	-0.038 (0.032)	0.000 (0.016)	0.029** (0.013)
BH q-value	0.613	0.470	0.969	0.482	0.988	0.159
Mean outcome of pre-ban cohorts	-.89	-.34	.25	.24	.093	.074
Observations	22,813	22,843	22,845	22,813	22,843	22,845

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the ban. All specifications use the weights described in section 3.3, calculated separately for each regression, and control for country-cohort fixed effects and subnational region fixed effects. “BH q-value” refers to the [Benjamini and Hochberg \(1995\)](#) q-value that adjusts for multiple hypothesis testing. Appendix Table A3 provides information about the availability of these variable by country.

C Full Sample Results

Our entire analysis – including that of women’s outcomes in Table 6 – restricts to mothers because our main outcome of interest is mortality of the next generation. In this section, we repeat the analysis of women’s outcomes without restricting to mothers, which offers a cleaner comparison to the Wilson (2022) results. Appendix Table C1 yields results that are similar but not identical to those reported in Wilson (2022). There are a number of reasons for the slight discrepancies. First, we use the most recent DHS for each country, which is more recent than the wave used in Wilson (2022) for about half of the countries. Second, we drop Kazakhstan (because of the small number of women in post-ban cohorts) and include the Gambia, whereas Wilson (2022) includes Kazakhstan but excludes the Gambia. Finally, we use the same treatment intensity variable used in our main analysis (which is calculated based only off of the restricted sample of mothers), whereas Wilson (2022) uses the full sample of women to calculate treatment intensity (though our results are similar when we do the same). Overall, our estimates are similar to the estimates in Table 4 of Wilson (2022), though for age at first marriage, age at first birth, and years of schooling, our estimates are slightly larger in magnitude, and for employment our estimates are slightly smaller in magnitude.

Comparing these results to those in Table 6, we find that estimates of the effects of child marriage bans on marriage and fertility-related outcomes are similar whether we restrict to mothers or not.¹⁸ However, this is not the case for schooling and employment. While child marriage bans significantly increase schooling and employment in the full sample of women, there are no significant effects among our sample of mothers. There are two potential reasons for this. First, it might be the case that the effects on schooling and employment were concentrated among younger women who have not yet given birth and are therefore excluded from our sample. Another possibility is that the results in Table 6 are picking up compositional changes due to selection into motherhood. The positive coefficient in column 1 of Table 2 suggests that there are “additional” post-ban mothers in high-intensity regions. If these additional mothers are less likely to be employed and have lower levels of schooling, this would result in an underestimate of the true effect of the bans on employment and schooling when we restrict only to mothers. Note that if this is the case, we are, if anything, underestimating the child mortality effects of child marriage bans, as discussed in section 4.5.

¹⁸Column 2 is, by construction, identical across the two tables because age at first birth is only available for mothers.

Table C1: Effects of Child Marriage Bans on Women's Outcomes (Full Sample of Women)

	(1)	(2)	(3)	(4)	(5)	(6)
	Age at first marriage	Age at first birth	Years of schooling	Currently employed	Age at first birth < 18	Total fertility
post-ban $\times$ intensity	1.07*** (0.11)	0.91*** (0.071)	0.51** (0.20)	0.066*** (0.010)	-0.032*** (0.0074)	-0.90*** (0.069)
Mean outcome of pre- ban cohorts	19.4	20.4	5.73	0.61	0.24	3.77
Observations	207,760	200,581	270,262	270,005	270,315	270,315

Notes: Standard errors clustered at subnational region level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. *postban* is an indicator variable coded as one if individuals are under the age of 18 at the time of the ban. All specifications control for country-cohort fixed effects and subnational region fixed effects. These regressions include all women, not just those who have ever given birth.